

KORALINK PROGRAM · AGENT RESOURCES

KoraLink App Agent User Manual

Your complete guide to selling, earning, and growing your business on the KoraLink platform

Version 2.0 , June 2026

Written for first-time Agent users

The Commons Project Foundation · TCP Africa · SFH Rwanda · ICT Chamber · Mastercard Foundation

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Before You Begin

This manual is written for first-time users of the KoraLink App. It walks you through every screen you will encounter as a KoraLink Agent - from logging in for the first time, to recording sales, ordering stock, using digital services, and managing your wallet.

You do not need prior experience with smartphones or business apps to follow this guide. Each section explains what you will see on screen, what each button does, and what to do next.

If you get stuck at any point, you can reach the KoraLink support team:

+250 796 602 807

Monday – Friday, 8:00 AM – 6:00 PM

What Is the KoraLink App?

The KoraLink App is your business tool as a KoraLink Agent. It runs on the mobile phone provided to you as part of the KoraLink Program. Through the app, you can:

- View your earnings and capital at a glance
- Record every sale you make to customers
- Check and manage your product inventory
- Place orders for new stock
- Sell digital services like airtime, electricity tokens, and more
- Track your voucher balance and repayment progress
- Receive messages and updates from SFH

Everything you do on the app is linked to your Agent account. Your sales, your stock, your earnings, and your repayments are all recorded automatically - you do not need to keep a separate notebook.

Understanding Your Account

Before exploring the app, it helps to understand a few key terms you will see throughout.

CAPITAL

The total value of everything your business holds - your stock and any earnings. When SFH gives you starter stock, that stock value becomes your starting capital.

VOUCHER

The startup support SFH provides to help you order stock without paying the full amount upfront. You receive vouchers as part of the KoraLink Revolving Fund. You repay this support gradually over time through your sales.

REPAYMENT

Happens automatically. Every time you pay for a stock order using your wallet, 4% of your wallet payment goes toward repaying your voucher balance. You do not need to calculate this yourself.

IKOFI

your in-app wallet. It is where your money is held for transactions on the platform. You will see this label on the fourth tab at the bottom of your screen.

COMMISSION

The profit you earn on each sale. For example, if a product costs RWF 550 and you sell it for RWF 750, you earn RWF 200 commission on that sale.

The Home Screen

When you open the KoraLink App and log in, the first screen you see is the **Home** screen. This is your daily starting point.

What You See at the Top

The orange bar at the top of the screen greets you by name - for example, "Hi, Perfect." This confirms you are logged into the correct account.

In the top-right corner, you will see three icons: a notification bell, a refresh button, and your profile avatar (a circle with your initial). Tapping the avatar opens your Profile page, covered in Part 5.

The Two Summary Cards

Just below the greeting are two cards sitting side by side.

The first card, on the left, shows your **Total Earnings** - the total profit you have made from sales on the platform. When you are just starting out, this will show RWF 0. It will grow as you record sales and earn commission.

The second card, on the right, shows your **Capital** - the total value of your business, including your stock. If SFH has loaded starter stock for you, you will see that value here. Tapping either card opens a more detailed breakdown.

The Voucher Repayment Banner

Below the summary cards is a white panel labelled **Voucher repayment**. This shows how much of your SFH startup support you have repaid so far.

You will see:

A small orange pill badge showing your repayment percentage (for example, "0% repaid")

A line showing your **Outstanding** balance - the amount still to be repaid

GOOD TO KNOW

This does not mean you owe a lump-sum payment right now. 4% of every wallet order you place goes toward this balance automatically. The outstanding amount decreases over time as you continue ordering and selling.

Tapping the arrow (>) on the right side of this banner takes you to your full repayment history, covered in Part 5.

Record Your Sales Section

The green section in the middle of the Home screen is where you start recording a sale. There are two buttons:

Record Sale (green button) - tap this when a customer buys a product from you. This opens the sales recording form, covered in full in Part 2.

View Sales History (white button) - tap this to see a list of all the sales you have previously recorded.

Below these buttons is the label "Earn rewards by logging sales." This is a reminder that recording your sales consistently contributes to your performance score and may make you eligible for additional incentive vouchers from SFH.

Need Help?

At the bottom of the Home screen is a blue panel labelled **Need Help? Check our FAQ for quick answers**. Tapping the arrow (>) takes you to the Frequently Asked Questions page, covered in Part 7.

The Bottom Navigation Bar

At the very bottom of the screen are four icons that stay visible no matter which screen you are on. These are your main navigation tabs:

- **Home** (house icon) - returns you to the Home screen
- **Inventory** (box icon) - shows your stock and lets you place orders
- **Services** (lightning bolt icon) - lets you sell digital services like airtime and electricity
- **Wallet** (wallet icon, labelled Ikofi) - your financial wallet for transactions

Recording a Sale

When a customer buys a product from you, you must record that sale in the app. This is how your commission is tracked and how your stock count is updated.

From the Home screen, tap the green **Record Sale** button.

1 Choose the Sale Type

At the top of the Record Sale screen, you will see two tabs: **Customer** and **Business**.

Select **Customer** if you are selling to an individual person - this is the most common type of sale. Select **Business** if you are selling to a company or organisation.

2 Select the Product

Below the tabs, you will see a search bar labelled "Search products..." and a list of **Available Products** from your current inventory.

Each product card shows:

- A photo of the product
- The product name and pack size
- The selling price (in orange - for example, 750 RWF)
- The cost price, the amount SFH charged you (for example, Cost: 550 RWF)
- Your current stock count (for example, Stock: 30)

Tap **View Details** on any product card to read more about it before selecting.

Check that the **Selling Price** field below the product list shows the correct amount - this field is editable. Use the **+** and **-** buttons in the Quantity field to set how many units the customer is buying.

Tap **+ Add Product** to add the item to the sale. The button stays grey until a product is selected and priced. You can add more than one product to a single sale by repeating this process.

3 Payment Details

Tap the **Payment Method** dropdown and select how the customer is paying. The most common option is Momo Code (MTN Mobile Money).

The **Notes** field is optional - use it for special instructions, such as a delivery request.

4 Customer Information

- **Customer Name** - enter the customer's full name
- **Phone Number** - the country code (+250) is pre-filled; enter the customer's number after it
- **Email** - optional; leave blank if not available

WHY THIS MATTERS

Capturing accurate customer information ensures the customer can receive an SMS receipt and that the sale is correctly attributed to your account.

5 Review and Submit

The **Sale Summary** panel shows your **Estimated Profit** (your commission) and the **Total Amount** the customer is paying. Review these figures before submitting.

When everything is correct, tap the orange **Record Sale** button at the bottom. Your stock count updates automatically. You can tap **View Sales History** at any time to review past sales.

Your Inventory

The **Inventory** tab (second icon in the bottom navigation bar) is where you see the products currently loaded on your account and place orders for new stock.

The Inventory Overview

At the top of the Inventory screen, two summary cards show:

- **Products** - the number of different products currently in your inventory
- **Inventory Value** - the total cost value of all your stock (for example, RWF 16,500)

My Requests

Just below the summary cards is a link labelled **My Requests**. Tapping this takes you to your **Stock Orders** screen, where you can see all the stock orders you have placed and their current status.

Your Product List

The main section of the Inventory screen lists all the products currently assigned to you. Each entry shows:

- A product image on the left
- The product name, category, and merchant (for example, Personal Care · SFH Rwanda)
- **Remaining Quantity** - how many units you still have in stock
- **Unit Price** - the selling price to customers
- **Commission** - how much profit you earn per unit sold

Use the search bar at the top to find a specific product quickly as your inventory grows.

Requesting New Stock

At the bottom of the Inventory screen is an orange **Request Product** button. Tap this when you want to order more stock from SFH or another merchant.

When you place an order, the system automatically checks your voucher eligibility and calculates how your order will be paid - splitting the amount between any available voucher balance and your Ikofi wallet. You do not need to manage this calculation yourself.

Each order you place goes through the following steps:

1. You select the products and quantities you want to order.
2. The system checks your voucher balance and determines how much voucher support can be applied.
3. The remaining amount is charged to your Ikofi wallet.
4. 4% of your wallet payment is automatically set aside as your revolving fund repayment.
5. The order is submitted for SFH approval.
6. Once approved, your stock is updated and you receive an SMS confirmation.

Reading Your Stock Orders

When you tap **My Requests**, you see a list of all your stock orders. At the top, filter buttons let you view orders by merchant (for example, SFH Rwanda or BASIL INDUSTRIES LTD).

Each order card shows the order reference number, merchant name, products ordered with quantities and values, total amount, a payment breakdown (Voucher used, Wallet charged, Loan repayment), the date and time created, and the current status.

Order statuses you may see:

- **Pending** - your order has been submitted and is waiting for SFH to review it
- **Approved** - SFH has approved the order and your stock is being prepared
- **Cancelled** - the order did not go through

IMPORTANT If an order is Cancelled, your wallet payment should be reversed automatically. If you see a cancelled order but your balance has not been restored, contact support with the order reference number.

Digital Services

The **Services** tab (lightning bolt icon, third in the bottom navigation bar) is where you sell digital services to customers. These services let you earn commission without needing physical products.

1 Select a Provider

When you tap the Services tab, you will first see the **Select Provider** screen, listing the digital service providers available on the platform.

Currently, the active provider is **DDIN**. Tap the DDIN card to access its services.

2 The DDIN Dashboard

After selecting DDIN, you are taken to the **DDIN Digital Services** screen. At the top, in the orange header bar, you can see your current **Digital Services Balance** - the float amount available to you for selling digital services.

You also have two buttons: **History** (view your full transaction history) and the **refresh icon** (update your balance).

Below the header is a list of **Available Services**:

- **Prepaid Electricity (EUCL)** - sell electricity tokens to customers with prepaid meters
- **MTN Airtime** - top up MTN mobile numbers for customers
- **Airtel Airtime** - top up Airtel mobile numbers for customers
- **RRA Tax** - help customers pay their Rwanda Revenue Authority tax obligations
- **Startimes PayTV** - renew Startimes satellite TV subscriptions for customers

Tap any service to begin a transaction for that service type.

How Digital Services Work

When a customer asks you to sell them airtime, electricity, or another digital service, you use your DDIN float balance to process the transaction. After the transaction is complete, the customer pays you in cash or via mobile money, and you earn a commission on the transaction.

Your DDIN float balance decreases with each transaction you process. When your float is running low, you can place a float top-up (refloating) order through the Inventory tab, the same way you order physical products.

Viewing Your Digital Services Transaction History

Tap the **History** button on the DDIN dashboard to see all your past digital service transactions.

The top of the History screen shows your total **DDIN Commission** earned. Below that, each transaction is listed with the service type, customer details and amount, date and time, and status (Successful or Failed).

For electricity transactions, a **Copy Token** button appears below the entry. Tap this to copy the customer's electricity token code so you can share it with them easily.

IMPORTANT

If a transaction shows a status of **Failed**, the customer was not charged and the service was not delivered. Do not ask the customer to pay for a failed transaction. Try again or contact support.

Your Profile and Ikofi Wallet

Accessing Your Profile

Tap the circular avatar icon in the top-right corner of the Home screen to open your **Profile** page.

Your profile shows your name, phone number, and the district you are assigned to (for example, Muhanga). This information is set during your SFH onboarding and cannot be changed from within the app. If any of your details are incorrect, contact your SFH supervisor.

Your profile has two tabs: **Overview** and **Business**.

Overview Tab

The Overview tab is the most important section of your profile for day-to-day financial awareness.

Voucher Repayment Card

This card shows you a full breakdown of your revolving fund repayment status:

- **Total Borrowed** - the total value of vouchers SFH has issued to you since you joined the program
- **Repaid So Far** - the total amount you have repaid through your orders to date
- **Outstanding** - the remaining balance you still need to repay

A progress bar beneath these figures fills up as your repayment percentage increases. The note "You repay 4% of every wallet order automatically" confirms the repayment rule.

Tap **View payment details** to expand this card and see a line-by-line history of every repayment deduction, with the date, amount, and source order for each entry. Each entry shows a status of either **Pending** (being processed) or **Completed** (confirmed).

Phone Repayment Card

This section tracks repayments related to any phone or device provided to you as part of your Agent equipment package. If this section shows an error, contact your SFH supervisor to confirm whether a device repayment is active on your account.

General Section

Below the repayment cards, three links give you access to important account features:

- **Manage Vouchers** - view all the vouchers issued to your account
- **Communications** - your in-app inbox for messages from SFH and the platform
- **Settings** - account preferences

At the bottom of the page is the **Logout** button. Always tap Logout when you finish using the app on a shared device.

Business Tab

Tap the **Business** tab at the top of the Profile screen to view your business registration details:

- **Business Name** - your registered trading name on the platform
- **Business Type** - this will show "KoraLink Agent" for all Agents in the program

This information is used when your sales records and commission reports are generated. If the business name is incorrect, notify your SFH supervisor.

Managing Your Vouchers

Tap **Manage Vouchers** from the General section of your profile to open the **Voucher** screen.

At the top, a summary card shows:

- **Voucher Balance** - any unused voucher credit currently available to you
- **Limit** - your remaining voucher ceiling (the maximum total voucher support you can still receive under the program)
- **Available to Request** - the amount currently available for your next order

Below the summary, the **My Vouchers** section lists every voucher that has been issued to your account, showing its unique voucher code, total value when issued remaining balance, and status (**Used** or **Active**).

A **Copy code** button appears under each voucher. You will not typically need to copy voucher codes yourself - the system applies them automatically when you place orders. If a support agent or SFH supervisor ever asks for your voucher code, this is where to find it.

The **Recent Requests** section at the bottom shows any standalone voucher requests you have submitted. Under the current system, vouchers are issued automatically when you place stock orders, so this section will typically remain empty.

The Ikofi Wallet (Wallet Tab)

Tap the **Wallet** icon (fourth tab in the bottom navigation bar) to access your **Ikofi** wallet.

Setting Up Your PIN

The very first time you tap the Wallet tab, you will see a screen that says "**PIN Not Set.**" This means your wallet is not yet secured.

You must set up a PIN before you can use any wallet features. Your PIN protects all financial transactions on the platform - including stock order payments, repayments, and digital service top-ups.

Tap the orange **Set Up PIN** button and follow the on-screen steps to create your PIN.

KEEP YOUR PIN SAFE

Choose a number you can remember but that others cannot easily guess. Do not share your PIN with anyone, including SFH staff or support agents. KoraLink support will never ask you for your PIN.

Once your PIN is set, tapping the Wallet tab will take you directly to your wallet dashboard. Every time you make a payment or financial transaction on the app, you will be asked to enter your PIN to confirm it. If you forget your PIN, contact KoraLink support immediately.

PART 6

Communications

Tap **Communications** from the General section of your profile to open your in-app inbox.

This is where the platform sends you important messages, including:

- Order confirmations after your stock orders are approved
- Repayment confirmation after a 4% deduction is processed
- Notifications from SFH about program updates
- Alerts about your account

At the top of the Communications screen, five filter tabs let you sort your messages:

All - shows every message

Unread - messages you have not opened yet

Read - messages you have already seen

Delivered - messages confirmed as delivered to your device

Failed - messages that could not be delivered

Pending - messages that are still being sent

TIP

If you expect to have received a message (for example, an order confirmation) but cannot see it, check the Failed and Pending tabs before contacting support.

Help and Support

Frequently Asked Questions

Tap the **Need Help?** banner on the Home screen to open the **Frequently Asked Questions** page.

At the top of the FAQ page, a search bar lets you type keywords to find specific answers quickly. Below it, the **Ask a Question** button lets you submit a question directly to the support team if you cannot find the answer in the FAQ list.

The FAQ is organised into categories:

- **Video Tutorials** - short instructional videos to help you learn the app features
- **Inventory** - answers about stock management, ordering, and product requests
- **Services** - help with digital services, DDIN transactions, and float
- **Ikofi** - guidance on your wallet, PIN setup, and financial transactions
- **Intelligra** - support for Intelligra digital services, available alongside DDIN

Tap any category to expand it and browse the questions and answers within.

Contacting the Support Team Directly

If you cannot find the answer you need in the FAQ, scroll to the bottom of the FAQ page to find the **Still Need Help?** section.

Support phone:

+250 796 602 807

Monday – Friday, 8:00 AM – 6:00 PM

Tap the orange **Contact Support** button to initiate a call or message directly from your device. You can also reach support through the KoraLink WhatsApp support line if you have been given that number by your SFH supervisor.

What to Do When...

I Want To...	Where to Go
Record a sale	Home → Record Sale → Select product → Enter details → Submit
Check my earnings	Home → tap Total Earnings card
See my stock	Bottom nav → Inventory tab
Order more stock	Inventory → Request Product
Check my order history	Inventory → My Requests
Sell airtime or electricity	Bottom nav → Services → Select DDIN
Check my repayment balance	Profile avatar → Overview tab → Voucher repayment card
Check my vouchers	Profile → General → Manage Vouchers
Set up my wallet PIN	Bottom nav → Wallet (Ikofi) → Set Up PIN
Get help	Home → Need Help? → FAQ, or call +250 796 602 807

REFERENCE

Glossary of Terms

Capital - The total value of your business, combining your stock value and earnings.

Commission - The profit you earn on a sale, calculated as the difference between the cost price and the selling price.

DDIN - The digital services provider currently active on the KoraLink platform, supplying airtime, electricity, tax payment, and PayTV services.

Float - Your digital services balance, used to process airtime, electricity, and other digital transactions for customers.

Ikofi - The Kinyarwanda name for your in-app wallet, where your money is held for platform transactions.

Intelligra - An additional digital service integration available on the KoraLink platform.

Outstanding Balance - The amount of voucher support you still need to repay.

Refloating - Topping up your digital services float balance through a stock order.

Repayment - The automatic 4% deduction applied to every wallet order, used to repay your voucher balance over time.

Voucher - Startup capital support provided by SFH that allows you to order stock without paying the full amount upfront.

Wallet Payment - The portion of a stock order paid from your Ikofi wallet, after any voucher balance has been applied.

Important Reminders

- Never share your PIN or login credentials with anyone. KoraLink support staff will never ask you for your PIN.
- Always record a sale in the app before the customer leaves. Sales recorded later may not match your stock accurately.
- If a digital service transaction fails, do not collect payment from the customer. Try again or contact support.
- If you see a cancelled stock order but your wallet balance has not been restored, contact support immediately with the order reference number.
- Your repayment balance decreases automatically each time you place a wallet order. Consistent ordering and selling is the most straightforward path to clearing your outstanding balance.

KoraLink

Agent User Manual · Version 1.0 · June 2026

For support: +250 796 602 807 | Monday – Friday, 8:00 AM – 6:00 PM

KoraLink Program - Powered by The Commons Project Foundation

In partnership with SFH Rwanda, ICT Chamber, and the Mastercard Foundation

KoraLink App

Agent Workflow Overview

| June 2026

TCP Africa · SFH Rwanda · ICT Chamber · Mastercard Foundation



KoraLink

KoraLink is a digital health commerce platform operating across Rwanda's districts. It equips community-based Agents with a mobile app, starter capital, and a product catalogue to sell health goods and digital services directly to consumers.

Key figures at a glance:

Agents

Field sales workforce deployed across Rwanda's districts

RWF 360,000

Maximum voucher allocation per Agent under the Revolving Fund

2+ Merchants

SFH Rwanda and BASIL Industries Ltd, with further partners on the roadmap

5 Services

Live digital service catalogue via DDIN integration

What Has Been Built

The following six modules are live in production as of now:

Agent Mobile App

Built on React Native and Expo. The Home, Inventory, Services, and Ikofi Wallet tabs are all live and accessible on Agent devices.

Stock Management

Full product catalogue, real-time inventory tracking, and multi-merchant order placement from a single interface.

Digital Services (DDIN)

Live integration with DDIN covering Prepaid Electricity (EUCL), MTN Airtime, Airtel Airtime, RRA Tax payment, and StarTimes PayTV subscription.

Ikofi Wallet

MVEND-powered wallet with mandatory PIN security, automatic voucher application at order placement, and 4% auto-repayment deduction on every wallet order.

Revolving Fund Engine

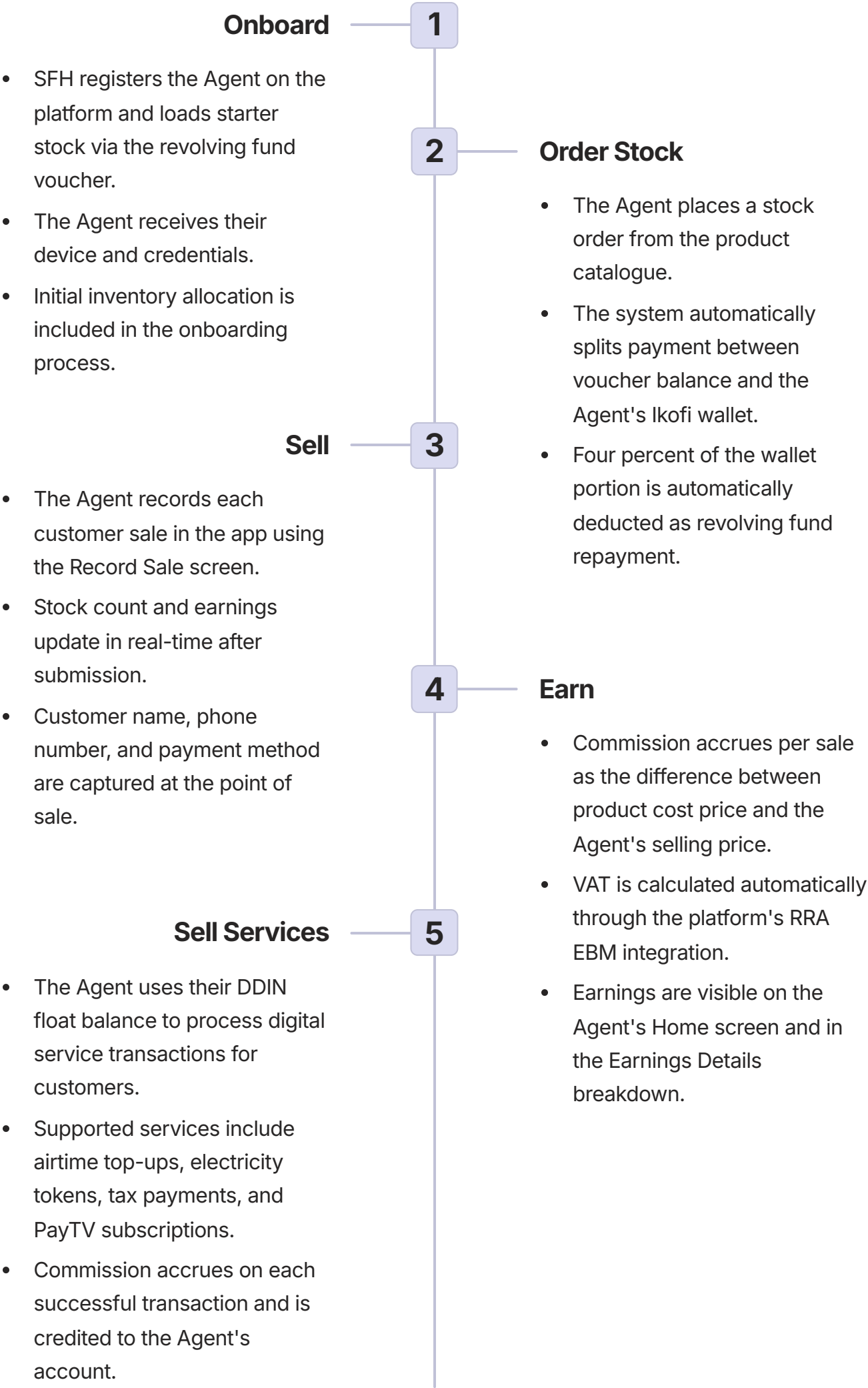
Two-phase voucher issuance model, real-time repayment tracking, and Agent-facing outstanding balance dashboard updated after every order.

Admin Platform

SFH admin dashboard for order approval, Agent performance oversight, and revolving fund monitoring.

The Agent Journey

The Agent workflow on the KoraLink platform follows five sequential stages:



📄 **Revolving Fund Loop:** 4% of every wallet order repays the Agent's outstanding voucher balance automatically. No manual action is required from the Agent or SFH admin.

Home Screen and Sales Recording

The Home screen is the Agent's daily starting point. It surfaces the Agent's key financial position and provides direct access to the two most frequent actions — recording a sale and checking sales history.

Live Capital View

- Total Earnings and Capital appear as summary cards at the top of every login.
- The Capital card reflects the total value of the Agent's current stock.
- Both cards are tappable and open detailed breakdowns for full visibility.

Voucher Repayment Progress

- A dedicated banner below the summary cards shows the outstanding balance.
- It also displays the repayment percentage in real time.
- The progress bar updates automatically after every order, without needing SFH support.

One-Screen Sale Recording

- Tapping Record Sale opens a single form for product selection, quantity, selling price, payment method, and customer information.
- A completed sale can be submitted in under 60 seconds, and stock updates automatically on submission.
- The form captures Customer Name, Phone Number (+250 prefix pre-filled), and optional Email for SMS receipts and sale attribution.

Inventory and Stock Orders

The Inventory tab gives Agents full visibility of their current stock and provides access to the stock ordering flow.

Stock Order Breakdown

Field	Value
Order Reference	#wimiqhq7
Merchant	SFH Rwanda
Products	2 items
Total Amount	RWF 23,712
Voucher Used	RWF 0
Wallet Charged	RWF 22,800
Loan Repayment (4%)	RWF 912
Status	Cancelled



Automatic 4% Repayment

Every wallet order deducts 4% for the revolving fund automatically, with no manual steps from the Agent or admin.



Multi-Merchant Ordering

Agents can order from SFH Rwanda, BASIL Industries Ltd, and future merchant partners from one catalogue using merchant filter tabs.



Order Audit Trail

Each order shows the voucher amount, wallet charge, and revolving fund repayment for Agents, admins, and auditors.

Digital Services via DDIN

The Services tab gives Agents access to the DDIN digital services catalogue using their DDIN float balance.

Digital Services Balance — Live Example

DDIN Float Balance: 760.00 Rwf

Available Services — Live Catalogue

Prepaid Electricity (EUCL)

Sell electricity tokens to customers with prepaid meters. Token is generated on transaction completion and displayed with a Copy Token button for immediate sharing with the customer.

MTN Airtime

Top up MTN mobile numbers on behalf of customers.

Airtel Airtime

Top up Airtel mobile numbers on behalf of customers.

RRA Tax

Facilitate Rwanda Revenue Authority tax payments for customers.

StarTimes PayTV

Renew StarTimes satellite TV subscriptions for customers.



Commission on Every Transaction

- RWF 2,057 total commission
- Accrues instantly on each successful service
- Reflected in the Agent's earnings



Token Delivery Built In

- EUCL token generated on completion
- Displayed inline with a Copy Token button
- Easy to share with the customer

Profile, Ikofi Wallet and Vouchers

Voucher Management

Field	Value
Voucher Balance	RWF 0
Voucher Limit	RWF 216,000
Available to Request	RWF 208,461.2
Total Borrowed	RWF 7,538.8
Repaid So Far	RWF 0
Outstanding	RWF 7,538.8

Three vouchers have been issued to this Agent:

- VCH654126459863 — RWF 742.8
- VCH640900885898 — RWF 2,476
- VCH134742802743 — RWF 4,320

These total RWF 7,538.8 and are all fully consumed. The Voucher Limit of RWF 216,000 reflects the remaining ceiling from the programme maximum of RWF 360,000.

 **Repayment Rule — displayed to Agent in the app:**
"You repay 4% of every wallet order automatically."

This plain-language rule is surfaced directly on the Agent's Profile screen. It provides a self-service explanation of the repayment mechanic, reducing inbound support queries to SFH without requiring the Agent to refer to documentation.

PIN-Secured Ikofi Wallet

- Wallet access is gated by a mandatory PIN setup on first access.
- Financial operations remain blocked until the PIN is configured.
- This includes stock order payments, repayments, and digital service payments.

Full Voucher History

- Every issued voucher appears in Manage Vouchers.
- Each entry shows the unique code, total value, remaining balance, and status.
- A Copy Code button is available for reference or support queries.

Business Profile

- The Business tab shows the Agent's registered Business Name and Business Type.
- Business Type is system-assigned as "KoraLink Agent" for all Agents.
- This classification governs access, voucher limits, commissions, and permissions.

Tech Stack and Live Integrations

Frontend

- React Native + Expo — Agent Mobile App
- Next.js — Web platform (Admin and Employer Dashboards)

Backend

- Next.js API Layer
- PostgreSQL Database
- MVEND — Wallet and payment gateway
- Twilio — SMS dispatch
- SendGrid — Email dispatch

Live Integrations

- DDIN — Digital services provider (airtime, electricity, tax, PayTV)
- IntelliGra — Phone repayment data for Agent Performance-Based Incentive Module
- RRA EBM — Rwanda Revenue Authority Electronic Billing Machine for VAT compliance
- ICT Chamber — Platform customisation, deployment, and ongoing operations

On the Roadmap

TeleHealth via Irembo · Health Insurance via EdenCare · USSD/SMS Offline Fallback for low-connectivity contexts · Resolvelt Ticketing Integration for Agent support



KoraLink Marketplace Platform Business Requirement Document (BRD)

Version 5.0

Date: January 2026

**Prepared For: Mastercard Foundation/ The Commons Project
Foundation / ICT Chamber**

Prepared By: Kevin Asimwe

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1. Introduction

This document details the definitive business requirements for the development, customization, and deployment of the Koralink Digital Platform. The platform is the core technological engine supporting the Digital Jobs for Youth in Health program (2024–2028) in Rwanda. This Program was established under an Agreement with the Mastercard Foundation (the "Foundation"). It is prepared by The Commons Project Foundation (TCP), in partnership with implementing organizations, including the ICT Chamber and Society for Family Health (SFH).

The Koralink Digital Platform, an evolution of the initial IHUZO e-marketplace foundation, serves as an OFFLINE to ONLINE business network designed to accelerate the digitalization of micro and small businesses and digital labor markets. Its primary mission is to advance education and relieve poverty for financially disadvantaged youth in Rwanda. The program objectives include creating 44,000 jobs in total, targeting approximately 14,640 direct digital



jobs in health, and enabling Koralink Agents to reach an average monthly earning of up to \$310 via the platform by the final year of the program.

The platform connects youth and women as Koralink Agents, Merchants/Admins (such as SFH), Employers, and Consumers. It provides the secure, scalable, and compliant digital infrastructure necessary to manage the complete Agent lifecycle. This lifecycle includes recruitment, blended e-learning, secure inventory management, real-time sales tracking, commission processing, and Revolving Fund repayment. Functionally, the system is built upon a modern technological stack, featuring a custom mobile application developed with React Native + Expo and utilizing a Next.js API backend with a PostgreSQL database (Prisma ORM), ensuring robust support for the integrated marketplace and financial services. The architecture is designed to maintain integrated offline registration and sales capabilities via USSD/SMS fallback for use in low-connectivity contexts.

2. Goals and Objectives

The platform's goals and objectives are centered on robust economic empowerment, agent enablement, and full compliance within Rwanda's digital health landscape.

Primary Aim

The core purpose of the KoraLink marketplace Koralink platform 3.0 is to advance education and relieve poverty for financially disadvantaged youth in Rwanda, predominantly women, by enabling them to become KoraLink agents and access work opportunities. Ultimately, the platform seeks to digitally enable Rwanda's youth, women, and rural communities to transact, learn, and earn safely, catalyzing sustainable economic and health impact, rooted in local partnerships.

Strategic Alignment and Focus

The Koralink marketplace platform is specifically positioned to complement Rwanda's Smart Village & Health Digitization strategies. The program emphasizes creating career ladders for earning opportunities and long-term digital health careers for unemployed youth. Koralink Agents are empowered as digital and commercial hubs by enabling them to sell essential goods and services and move up through these career ladders. The progression model is designed to be tracked, mentored, and supported, moving From Level A (survival income) → To Level C (multi-service offering) → To Level F (small business employer).

Key Objectives

The platform is designed to support the following specific, measurable objectives:

- **Economic Empowerment:** To relieve poverty by enabling the creation of an estimated 14,640 direct and 29,360 indirect work opportunities. This includes ensuring Koralink Agents reach an average monthly earning of up to \$310 via the Marketplace Platform by the final year of the program.
- **Facilitating Commerce:** To facilitate the sales of health goods and services by KoraLink agents in their communities, including managing inventory, collecting payments, and processing commissions. It must also support Merchants/Admins (like SFH) with real-time inventory, order, and finance tools, enabling multi-product marketplaces and fund management options.
- **Skills Development:** To implement learning and certification tracks for Agents by providing e-training and performance management tools to complement in-person training. This enables Agents to develop professional skills, earn certifications, and track their progress and earnings for potential employers.
- **Livelihoods & Jobs:** To create a platform for upskilled KoraLink agents to offer their skills/expertise for hire. This links them with prospective employers and allows employers to post, manage, and fulfill jobs using a trusted pool of accredited local Agents.
- **Financial Sustainability:** To enable the establishment and utilization of a Revolving Sustainability Fund through Repayable Grants to Program participants for initial goods

and devices, with repayments reinvested. This also establishes sustainable, diversified income streams for KoraLink agents that extend beyond the initial 4-year project window.

- **Compliance and Traceability:** To Deliver full compliance, traceability (audit logs), and privacy using strong identity, role, and consent flows for all users.
- **Accessibility:** To provide integrated offline registration/sales via USSD/SMS fallback for low-connectivity scenarios.

3 Out-of-Scope

The Koralink Digital Platform, as the core technological engine, maintains a defined boundary concerning its direct responsibilities and functional capabilities. The platform is not directly responsible for or does not include the following operational or functional areas:

Physical Logistics and Distribution: The platform provides digital tracking and management tools, such as the Dispatch Management System, but it does not handle the physical movement of products or the logistics fulfillment executed by external partners. Physical distribution of health goods to KoraLink Agents is primarily handled by the implementing partner, Society for Family Health (SFH).

Direct Healthcare or Diagnostic Services: The system's scope is to enable the sale and distribution of health goods and services (e.g., telehealth subscriptions) and to train Agents to provide these services. However, the platform does not execute the actual medical or diagnostic service provision itself.

Comprehensive Financial Management Outside Transactions: The platform is not responsible for overall comprehensive financial management and procurement activities that occur *outside* of its direct transaction processing, such as central budget oversight. These activities may be managed by a dedicated Fund Manager if one is retained by the Mastercard Foundation.

Program Governance and Contract Management: The platform does not manage the overall contract or high-level administration between the Foundation and The Commons Project Foundation (the Organization); these matters are governed by the main Agreement.

Experimental Integrations and Future Enhancements (Current Phase): Third-party or experimental integrations are deferred to future phases of the platform's development. Specifically, the platform is not currently tasked with implementing:

- A dedicated Customer Relationship Management (CRM) system specifically for service delivery.
- The capability for advanced KoraLink Agents to track installed Internet of Things (IoT) devices and data collected by those devices.

4. Stakeholders

The Koralink Digital Platform supports a diverse ecosystem of stakeholders, each with specific roles, responsibilities, and access levels, crucial for the success of the Digital Jobs for Youth in Health program. The platform is designed to manage the interactions between these roles through granular access control and defined workflows.

Primary Stakeholders and Funders

- **Mastercard Foundation:** Provides funding and oversight for the Program, defines the charitable purposes and Program objectives, and receives mandated reports. The Foundation may also retain a **Fund Manager**.
- **The Commons Project Foundation (TCP):** The Organization responsible for performing the Services under the Agreement. TCP manages the Revolving Fund, oversees overall Program delivery, subcontracts implementing partners, and is responsible for maintaining all necessary records. As the platform's operator, TCP is an independent data controller regarding the personal data processed.

Implementing and Operational Partners

- **ICT Chamber:** An implementing partner responsible for the development, deployment, operation, and maintenance of the Koralink marketplace platform, including providing necessary technical support. ICT Chamber is involved in platform services and may receive disbursements as a subcontractor.
- **Society for Family Health (SFH), Rwanda:** A key implementing partner and the primary **Merchant/ADMIN** on the platform. SFH is responsible for:
 - **Koralink Agent recruitment, training, onboarding, and management.**
 - **Distribution of health products** to KoraLink Agents and handling physical logistics.
 - **Overseeing product assignment** for the Revolving Fund.
 - **Monitoring and Evaluation (M&E)**, including tracking long-term outcomes for Agents.
 - **SFH Staff/Supervisors** are users of the system requiring tools for enhanced monitoring, approval/rejection of Agent applications, and visibility into Agent performance.

Platform Users and Beneficiaries

- **KoraLink Agents** The primary beneficiaries and field operators of the platform, targeted for economic empowerment, including youth and women. Their roles include selling goods/services, undergoing e-training, seeking jobs, managing their own stock, and recording sales.
 - Agent Candidate Categories range from Level A (no prior experience, motivated) to Group D (professionals like Nurses offering higher-value services).
- **Consumers (Customers):** End-users who purchase health goods and services listed on the marketplace from KoraLink Agents. Their key actions include browsing products, placing orders, and making digital payments.



- **Employers:** Businesses and organizations that post verified job opportunities and skills requirements, review upskilled KoraLink Agents, and facilitate hiring via the platform's Job Marketplace.

Specialized and Governance Roles

The platform supports specialized user roles for management and compliance:

- **SUPER_ADMIN:** System owner with full access and oversight, responsible for system configuration and user management.
- **ADMIN/Merchant:** Program operators like SFH who manage products, inventory, funds, application review, and overall operational administration.
- **Interviewer:** A specific role for staff (like SFH) who manage candidate interviews, scoring, and onboarding approvals.
- **Application Agent:** A role supporting application management and basic reporting.
- **Health Goods and Services Providers:** Organizations (like SFH) and manufacturers (like TeleHealth/Irembo, Health Insurance/EdenCare) whose products are listed and sold by KoraLink Agents.
- **Government Entities:** Involved in policy alignment, training development (e.g., Data Protection Officer training), and potential integration or endorsement, including the **Ministry of Health, National Cyber Security Agency (NCSA)**, and National Health Intelligence Center. The platform also requires alignment with the Rwanda Revenue Authority (RRA) for EBM tax compliance.
- **Fund Manager (if retained):** A service provider appointed by the Foundation to assist with financial management, procurement, and administration related to the Program and Revolving Fund.

Future and Noted Stakeholders

Potential future stakeholders include individual enterprises/sole proprietors (KoraLink Agents who formalize their businesses) and Healthposts and regional offices that could serve as future wholesalers.

5. User Requirements

This section details the specific needs and expectations of the various user roles, prioritizing seamless operation, economic empowerment, and robust oversight within the Koralink Digital Platform ecosystem.

KoraLink Agents: Core Operational & Financial Empowerment

KoraLink Agents, as the primary field operators and beneficiaries, require a secure, intuitive, and accessible platform to manage their livelihoods and professional development.

- **Real-time Business Management:** Agents need a user-friendly interface to track their available inventory ("My Stock Dashboard"), monitor sales, and view their current commission earnings (dashboards). They must be notified when stock is assigned to them by administrators.
- **Financial Tracking:** Agents require real-time visibility of their Revolving Fund repayment status and the ability to view commission earnings and manage potential Digital Financial Services (DFS) accounts linked via the platform. The architecture is designed so that all consumer sales are unambiguously credited to a single Agent's digital storefront, ensuring 100% of consumer sales are tracked and attributed to them for commission and inventory management.
- **Performance and Incentive Visibility:** Agents require real-time access to their monthly performance score breakdown across all three components (Repayments, Sales, Orders), their current tier status (Bronze / Silver / Gold / Platinum), quarter-to-date eligibility progress against the quarterly repayment compliance threshold, and a historical record of incentive vouchers received.
- **Workforce Planning:** Agents need practical tools, such as the Task Tracker or Work Week Planner, to effectively manage their time, allocate effort across sales, gigs, and

training, and estimate potential income streams. Agents must also have a clear way to indicate their unavailability ("offline" status) for work or sales activities, which may require supervisor approval.

- **Professional Identity:** Agents need tools to create and update their professional profile, showcasing qualifications, certifications, and performance metrics (sales volume, commission history, repayment rates) to employers and administrators. They also require copies of their signed contracts and a form of identification to enhance their credibility in their communities.
- **Accessibility:** Agents require reliable access to the platform via provided devices and data. Crucially, the system must support full offline use for rural users, including registration and sales capabilities via USSD/SMS fallback for low-connectivity contexts.

Operational Staff (SFH Admins/Supervisors): Oversight and Management

SFH staff, who manage Agent recruitment, logistics, and performance, require advanced tools to ensure program compliance and efficiency.

- **Recruitment & Onboarding:** Supervisors require enhanced monitoring tools, including visibility into KoraLink Agent availability status, to manage and support their assigned Agents effectively. SFH Admins need tools to approve/reject KoraLink Agent registrations to ensure only trained candidates join the platform.
- **Performance and Financial Oversight:** Admins need centralized dashboards providing end-to-end visibility into Agent performance, inventory velocity, and fund distribution. They require specific tracking tools to monitor the marketplace economics, including the price Agents purchase products for (acquisition value) versus the price they sell for (sales value).
- **Fund Management:** Admins require the ability to filter Agents by repayment delinquency to prioritize support. They need tools to manage hardship cases, authorize partial payments, and perform bulk actions related to fund distribution.

- **Impact Tracking:** SFH, responsible for Monitoring and Evaluation (M&E), requires the ability to track long-term program outcomes beyond the platform, such as whether dropouts successfully secure jobs outside the Koralink ecosystem.

Ecosystem Partners (Consumers & Employers)

The platform must provide streamlined interfaces for external partners interacting with KoraLink Agents and the marketplace.

- **Consumers (Customers):** Consumers require a simple way to browse and search for health goods and services offered by local Agents, place orders, make digital payments, and receive timely order confirmations and updates. The consumer experience requires straightforward product discovery and readily available customer support.
- **Employers:** Employers need a portal to create company profiles, post verified job openings, browse the profiles and qualifications of upskilled Agents, and submit reviews/ratings for hired Agents. They require identity-secured hiring and vetting workflows based on Agent certifications and trackable performance metrics to find qualified workers.

Platform Management (Organization/Funders)

Staff from TCP (the Organization) and the Mastercard Foundation (or Fund Manager) require high-level visibility and audit capabilities.

- **Reporting and Analytics:** Organization/Foundation Staff need access to comprehensive reports (financial, narrative, KPI performance), dashboards summarizing Program status, and tools for data analytics. They require tools to manage all listings and postings, including training content, product listings, and job postings.
- **Financial Auditing:** If a Fund Manager is retained, they require access to financial data related to transactions, disbursements, fund balances, and repayable grants facilitated or tracked by the platform, along with tools to generate specific financial reports.

- **Compliance Oversight:** Platform managers need tools to oversee content and monitor compliance across the system.

6. Functional Requirements

6.1. User Management & Authentication Module

This foundational module governs secure digital identity verification and access control, ensuring compliance with Rwanda's data protection laws.

Secure Registration & Access

Purpose: To enable secure onboarding and authentication for all user types (Agents, Consumers, Employers, Admins) while ensuring data privacy.

Key Flow: Registration → Identity Verification (SFH-verified ID check) → Role-Based Access → Authentication → Session Management.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to register using my login credentials so I can access the platform quickly.

Acceptance Criteria: SFH ID auto-validates via API.

User Story: As a Consumer, I want to sign up with just my phone number so I can browse products fast.

Acceptance Criteria: OTP sent via SMS for verification.

Profile Management

Purpose: To centralize KoraLink Agent professional data (skills, certifications, earnings, ratings) to enable job matching and transparency.

Key Flow: Auto-Generated Profile Creation (SFH-verified data) → Agent edits personal info/uploads certificates → Employer views profile/ratings.

User Stories & Acceptance Criteria:

User Story: As a newly registered KoraLink Agent, I want my profile auto-created with SFH-verified data (name, ID, training) so I don't need manual entry.

Acceptance Criteria: Mandatory fields are SFH ID, assigned location, and certifications.

Non-Functional Requirements

- Authentication must utilize JWT token-based authentication.
- Registration must include Consent Checkboxes for PII processing, adhering to Rwanda's data protection laws.
- The system must maintain complete role-history auditability.

6.2. KoraLink Agent Application Management Module

This module manages the integrated workflow for recruitment, scoring, and formal onboarding of Agent candidates, supporting SFH's role in eligibility assessment.

Application & Interview Management

Purpose: To support inclusive recruitment, eligibility assessment, and program onboarding for youth, particularly women in rural areas.

Key Flow: Candidate submits application (mobile/web/USSD) → System assigns preliminary vulnerability group (A/B/C) → Interviewer records interview outcome/scoring → Approval/Rejection → Status upgrade to KoraLink Agent.

User Stories & Acceptance Criteria:

User Story: As an SFH Admin, I want to approve/reject KoraLink Agent registrations to ensure only trained candidates join.

Acceptance Criteria: Dashboard shows pending registrations; Rejection sends SMS with reason.

User Story: As an SFH Admin, I want to view KoraLink Agent applicants from my district and filter them by eligibility criteria.

Acceptance Criteria: Admin can view/filter only their assigned region.

Scoring System

Purpose: To utilize vulnerability and readiness indicators for objective candidate assessment.

Key Flow: Application form captures vulnerability indicators (income, disability, digital access)

→ Scoring system processes inputs (confidence, device verification) → Result determines application status.

User Stories & Acceptance Criteria:

User Story: As the platform, I want to send SMS login info and training links/info to approved KoraLink agents.

Acceptance Criteria: Upon approval, the user's role is upgraded to KoraLink Agent and a confirmation SMS is sent.

Non-Functional Requirements

- The module must integrate with the existing OTP authentication and user role engine.
- Personal and sensitive data must be stored securely in compliance with Rwanda's data protection laws.

6.3. Stock Management & Inventory Module

This module provides digital tracking and control over product inventory, ensuring full traceability with unit-level granularity required by the program's centralized fulfillment model.

Inventory Control and Allocation

Purpose: To manage stock levels, prevent KoraLink Agents from requesting out-of-stock items, and ensure real-time inventory visibility.

Key Flow: Super User/Warehouse Staff assign stock to Agent Inventory (special branch) → Agent views assigned stock → Agent records sale → Inventory auto-deducts in real-time.

User Stories & Acceptance Criteria:

User Story: As an SFH Admin, I want to be notified for replenishment when stock falls below a configured threshold.

Acceptance Criteria: Automated alerts trigger when stock falls below custom thresholds.

User Story: As a KoraLink Agent, I want low-stock SMS alerts when inventory falls below 5 units.

Acceptance Criteria: SMS format: “Low stock! Only 3 nets left. Reply RESTOCK to request more”.

Audit Trail

Purpose: To maintain an unchangeable record of all product movements to satisfy funder and regulatory audit requirements.

Key Flow: Stock moves from warehouse to Agent → System logs transfer source, destination, initiator, and status.

User Stories & Acceptance Criteria:

User Story: As the System, I must maintain an immutable audit trail for all product movements.

Acceptance Criteria: Audit log records transfer source, destination, initiator, status, and quantity with unit-level granularity.

Non-Functional Requirements

- Granular access permissions must limit Agents to viewing their own stock, orders, and commissions.
- All inventory movements must be logged and archived for the 10-year fiscal retention period.

6.4. Sales & Commission Module

This core transactional module records all sales, automates commission calculation, and ensures mandated RRA EBM tax compliance.

Sales Recording & Tax Compliance

Purpose: To track Agent sales and ensure the platform adheres to RRA mandates by submitting required transactional data in real-time.

Key Flow: Agent records sale → Inventory auto-deduction → Commission calculated (percentage-based) → Invoice auto-submission to RRA EBM.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to see commission breakdowns per sale to verify accuracy.

Acceptance Criteria: Tapping any transaction displays the calculation details (e.g., “RWF 7,000 sale @ 10% = RWF 700”).

User Story: As a consumer, I want to pay via mobile money (USSD/STK push/QR supported by Mwend), so that I can complete my purchase securely.

Acceptance Criteria: Customers receive real-time payment confirmation.

Compliance Workflow

Purpose: To ensure all sales are fiscally compliant and prevent legal risks.

Key Flow: Invoice generated → System performs TIN Validation for Employers → Successful API call to RRA → Failure triggers VSDC fallback mechanism.

User Stories & Acceptance Criteria:

User Story: As the System, I must auto-submit all invoices generated in the backend engine to RRA.

Acceptance Criteria: Failed API calls must be logged and allow for retrying or escalation procedures.

Non-Functional Requirements

- Requires official RRA EBM certification before the integration can go live.
- The system must implement the VSDC fallback protocol to log and retry failed invoice submissions immediately upon connectivity restoration.

6.5 Wallet & Financial Management Module

This module manages the Revolving Fund and all financial flows via the integrated third-party wallet system (MVEND). It governs how KoraLink Agents receive startup capital, how that capital is disbursed as vouchers during order placement, and how repayments are progressively collected to sustain the fund.

Voucher-Based Revolving Fund Model Overview

Purpose: To provide KoraLink Agents with startup capital (the Maximum Allowed Voucher) disbursed as product vouchers during order placement, subject to a two-phase issuance system that balances Agent access to inventory with progressive repayment of the Revolving Fund.

Vouchers are no longer requested independently and approved by SFH before ordering. Under the revised model, voucher allocation is computed by the system during order placement and is embedded directly into the order approval workflow. This eliminates the separate voucher-request queue and ties voucher issuance to actual inventory needs.

Key Flow: Agent places order → System checks voucher eligibility → Applies unused voucher balance → Determines phase (Unrestricted or Restricted) → Computes voucher allocation and wallet payment split → Applies Revolving Fund repayment percentage → Order submitted with voucher request for approval.

System Constants (Configurable Per Agent)

The following constants govern the voucher issuance engine. All values are configurable at the system level and may be overridden per Agent where program rules permit.

Constant	Default Value	Description
Maximum Allowed Voucher	360,000 RWF	Total cumulative voucher ceiling per Agent. Agent cannot receive vouchers exceeding this amount across their entire program lifecycle.

Initial Unrestricted Ceiling	50,000 RWF	Voucher amount available without repayment enforcement. Orders within this ceiling do not require minimum wallet repayment.
Maximum Voucher per Request	50,000 RWF	Cap on new voucher allocation in a single order transaction.
Wallet/Voucher Contribution Ratio	70% ^d	In the Restricted Phase, the Agent must contribute a minimum of 70% of their Net Outstanding Voucher via wallet before a new voucher is issued.
Revolving Fund Repayment Rate (Products)	4%	Percentage applied to total wallet payment to contribute toward Revolving Fund repayment on product orders.
Revolving Fund Repayment Rate (Services)	4%	Percentage applied to total wallet payment to contribute toward Revolving Fund repayment on service orders.

Core Definitions

The following terms are used throughout the voucher issuance and repayment logic. These definitions supersede any conflicting terminology from BRD v4.0.

Term	Definition	Formula
Gross Cumulative Received Voucher	Total value of all vouchers ever issued to the Agent across their program lifecycle.	—
Total Repaid Startup Capital	Cumulative amount the Agent has repaid through the Revolving Fund repayment mechanism (the 4% charges on wallet payments).	—
Net Outstanding Voucher	The Agent's current unpaid voucher balance, excluding unused voucher amounts.	Gross Received – Repaid – Unused
Unused Voucher Balance	Previously issued voucher amount that has not yet been applied to any order.	—
Voucher Ceiling Amount Balance	Remaining capacity before the Agent hits their Maximum Allowed Voucher.	Max Allowed – Gross Received
Gross Order Value	Full order amount before deducting any unused voucher balance.	—

Net Order Value	Order amount after deducting any applicable unused voucher balance.	Gross Order – Applied Unused
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Mandatory Guiding Principles

The following principles are non-negotiable constraints governing the voucher issuance engine:

- **Order-Integrated Issuance:** Vouchers are issued only during order placement. Agents cannot request vouchers independently outside of an active order. Additional vouchers function as wallet top-ups within the order flow.
- **Ceiling Enforcement:** The total value of vouchers issued to any Agent must never exceed the Maximum Allowed Voucher (currently 360,000 RWF). The system must block any allocation that would breach this ceiling.
- **Per-Request Cap:** The voucher issued within a single transaction must not exceed the Maximum Voucher per Request (currently 50,000 RWF).
- **Voucher-Before-Order Approval:** When an order includes a voucher request, the voucher must be approved first, then the order. If the voucher cannot be approved, the order must be cancelled. These are not independent approval flows.
- **Product Eligibility Segregation:** Products that accept voucher payment cannot be mixed with products that do not in a single order. If any product in the order does not accept voucher payment, the entire order must be paid via wallet.
- **Revolving Fund on All Wallet Payments:** All wallet payments are subject to the Revolving Fund repayment percentage (currently 4%), regardless of whether the order involved a voucher. This charge is capped at the Agent's remaining outstanding startup capital.
- **Digital Services Scope:** The 360,000 RWF Maximum Allowed Voucher ceiling is shared across both physical product orders and digital service float orders (Float and Refloating). Voucher issuance logic applies equally to float and refloating orders placed through the marketplace, subject to the digital service provider accepting voucher payment via their API integration with the platform. SFH deposits the startup pack into the SFH Master Account on the digital service provider platform; the marketplace then disburses the corresponding vouchers to Agents via APIs provided by the digital service providers, following the same eligibility logic and phase determination applied to physical product orders. If a digital service provider does not accept voucher payment, the full order amount is charged to the Agent's wallet and the Revolving Fund repayment percentage still applies. The shared 360,000 RWF ceiling is consumed regardless of whether the order is for physical products or digital services.
- **System-Determined Allocation:** Voucher amounts are computed by the system based on defined constraints. Agents do not directly control the voucher amount issued; they place orders, and the system determines the optimal voucher/wallet split.

Voucher Eligibility & Issuance Logic

The voucher issuance engine follows a deterministic, multi-step process for every order placed by a KoraLink Agent. The system evaluates eligibility, applies existing balances, determines the applicable phase, and computes the final payment split.

Step 1: Check Voucher Eligibility

Purpose: To determine whether the current order qualifies for voucher payment based on product eligibility.

- **If all products in the order accept voucher payment** → Proceed to Step 2 (voucher logic).
- **If any product does not accept voucher payment** → No voucher is applied. The full order amount is charged to the Agent's wallet. Revolving Fund repayment still applies (see Step 6).

Step 2: Apply Existing (Unused) Voucher Balance

Purpose: To consume any previously issued but unused voucher balance before considering new voucher issuance.

Any unused voucher balance must be fully utilized before the system issues a new voucher. The applied amount is the lesser of the order amount and the unused balance:

$$\text{Applied Unused Voucher} = \min(\text{Order Amount}, \text{Unused Voucher Balance})$$

$$\text{Remaining Order} = \text{Order Amount} - \text{Applied Unused Voucher}$$

If the remaining order amount is zero after applying the unused voucher, no new voucher is issued and the order proceeds directly to the order summary. No wallet payment is required beyond the Revolving Fund repayment percentage.

Step 3: Determine Phase

Purpose: To determine whether the Agent is operating in the Unrestricted Phase or the Restricted Phase, which governs the repayment enforcement rules.

Phase	Condition	Behavior
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Unrestricted	Gross Cumulative Received Voucher < Initial Unrestricted Ceiling (50,000 RWF)	Vouchers are allocated freely to cover the order. No minimum wallet repayment is enforced. The system allocates up to the remaining unrestricted limit, the per-request cap, or the remaining total voucher capacity — whichever is lowest.
Restricted	Gross Cumulative Received Voucher $\geq$ Initial Unrestricted Ceiling (50,000 RWF)	The Agent must contribute a minimum wallet repayment (70% of Net Outstanding Voucher) before any new voucher is issued. New vouchers are allocated only after repayment enforcement.

Split-Phase Orders

A single order may span both phases. If an Agent has 40,000 RWF in cumulative vouchers and places an order for 80,000 RWF, the system first allocates the remaining 10,000 RWF under the Unrestricted Phase, then processes the remaining 70,000 RWF under Restricted Phase conditions (including repayment enforcement and voucher limits).

Step 4: Unrestricted Phase Allocation

Purpose: To allocate vouchers within the initial unrestricted ceiling without requiring repayment enforcement.

The system computes the unrestricted voucher allocation as the minimum of four values:

- The remaining order amount (after applying unused voucher)
- The Maximum Voucher per Request (currently 50,000 RWF)
- The remaining unrestricted limit (Initial Unrestricted Ceiling – Gross Cumulative Received Voucher)
- The remaining total voucher capacity (Maximum Allowed Voucher – Gross Cumulative Received Voucher)

Unrestricted Voucher = min(Remaining Order, Max Per Request, Remaining Unrestricted, Remaining Capacity)

If the unrestricted allocation fully covers the remaining order, the order proceeds to the summary. If a balance remains, the system enters the Restricted Phase for the remainder.

Step 5: Restricted Phase Allocation

Purpose: To enforce minimum repayment and allocate additional vouchers under tighter constraints.

Compute Net Outstanding Voucher:

$$\text{Net Outstanding} = \text{Gross Cumulative Received} - \text{Total Repaid} - \text{Unused Voucher Balance}$$

Enforce Minimum Required Repayment:

The Agent must contribute a minimum wallet repayment equal to the Wallet/Voucher Contribution Ratio (currently 70%) of the Net Outstanding Voucher. This amount is capped at the remaining order amount — the system never requires the Agent to pay more than what the order costs:

$$\text{Required Repayment} = \min(70\% \times \text{Net Outstanding}, \text{Remaining Order After Unrestricted})$$

Allocate Restricted Voucher:

After applying the required repayment to the remaining order, the system computes the restricted voucher allocation as the minimum of:

- The remaining order amount after repayment
- The Maximum Voucher per Request (50,000 RWF)
- The remaining total voucher capacity (adjusted for any unrestricted allocation already issued in this transaction)

$$\text{Restricted Voucher} = \min(\text{Remaining After Repayment}, \text{Max Per Request}, \text{Remaining Capacity} - \text{Unrestricted Voucher})$$

Step 6: Compute Wallet Payment & Revolving Fund Repayment

Purpose: To determine the total wallet charge including the mandatory Revolving Fund contribution.

The wallet payment covers any portion of the order not covered by vouchers (unused + unrestricted + restricted):

$$\text{Wallet Payment} = \text{Order Amount} - (\text{Applied Unused} + \text{Unrestricted Voucher} + \text{Restricted Voucher})$$

The Revolving Fund repayment is then applied as a percentage of the wallet payment:

$$\text{Revolving Fund Repayment} = \text{Wallet Payment} \times 4\%$$

Repayment Cap (Critical Business Rule)

The Revolving Fund repayment is capped at the Agent's remaining outstanding startup capital (Gross Cumulative Received Voucher – Total Repaid). Once the Agent has fully repaid their total voucher received, no further repayment charges are applied. This prevents overpayment and ensures the fund operates as a true revolving mechanism.

The total amount deducted from the Agent's MVEND wallet is:

$$\text{Total Wallet Payment} = \text{Wallet Payment} + \text{Revolving Fund Repayment}$$

Step 7: Order Reconciliation

The system must validate the following reconciliation identity before submitting the order:

$$\text{Order Amount} = \text{Total Voucher Used} + \text{Wallet Payment}$$

Where Total Voucher Used = Applied Unused + Unrestricted Voucher + Restricted Voucher. If this identity does not hold, the order must not proceed and the system must log the discrepancy for investigation.

Voucher Eligibility & Order Flow: Decision Diagram

The following diagram illustrates the complete decision flow executed by the system during order placement. It maps directly to Steps 1–7 described above, showing how the system determines voucher eligibility, applies existing balances, transitions between the Unrestricted and Restricted phases, enforces minimum repayment, and computes the final payment split.

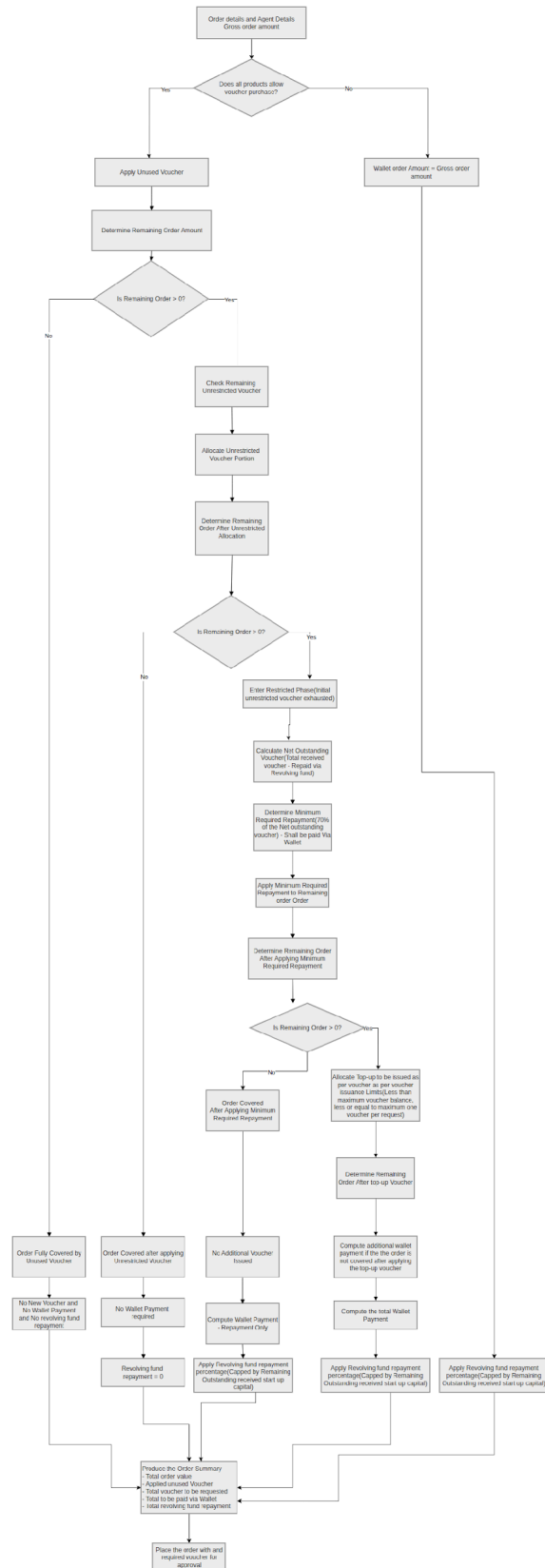


Figure 1: Revised Voucher Eligibility & Order Placement Flow

Standalone Revolving Fund Repayment

Purpose: To allow KoraLink Agents to make voluntary repayments toward their Revolving Fund balance outside of the order placement flow.

Key Flow: Agent initiates repayment (via app) → System computes Net Outstanding → Agent enters repayment amount → System validates amount $\leq$ Net Outstanding → Accept or Reject.

The platform supports voluntary repayments through two channels:

- **Mobile App:** The Agent selects “Make Custom Repayment” from the Wallet section, enters any amount up to their total outstanding balance, and the system uses MVEND wallet integration to process the payment in real-time.
- **USSD:** The Agent accesses the repayment menu via USSD shortcode, enters the desired amount, and the system processes the payment with SMS confirmation.

Validation Rule: The repayment amount must not exceed the Agent’s Net Outstanding Voucher. If the entered amount exceeds the outstanding balance, the transaction is rejected and the Agent is notified of the maximum allowable repayment.

Upon acceptance, the system must immediately recalculate the remaining principal, update the Agent’s progress bar dashboard, and send an automated SMS notification confirming the payment and stating the new remaining balance.

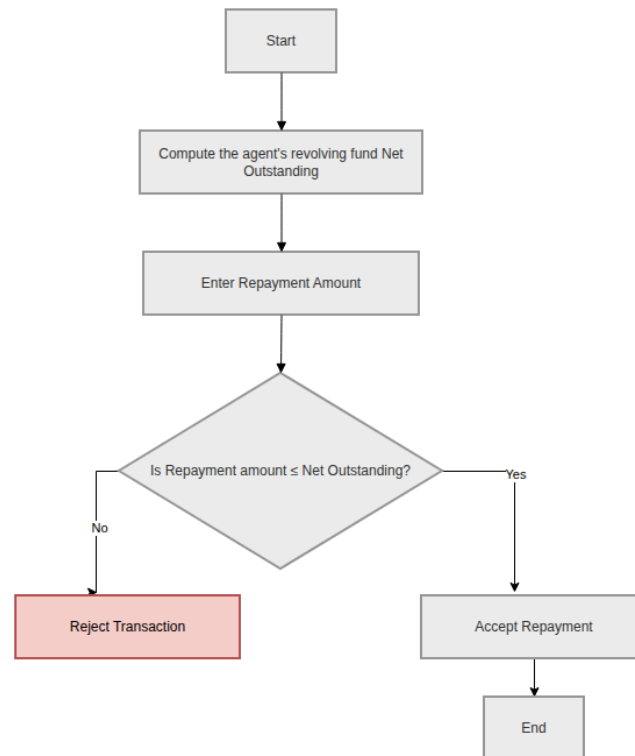


Figure 2: Standalone Revolving Fund Repayment Validation Flow

User Stories & Acceptance Criteria

Agent: Order with Voucher

User Story: As a KoraLink Agent, I want the system to automatically determine the best voucher/wallet split when I place an order, so that I can restock efficiently without manually calculating my eligibility.

Acceptance Criteria:

- The order screen displays the Agent's current Voucher Ceiling Balance (Maximum Allowed – Gross Cumulative Received).
- If all products accept vouchers, the system auto-computes the voucher allocation following the Unrestricted/Restricted phase logic.
- If any product does not accept a voucher, the system blocks voucher allocation and displays a clear message indicating full wallet payment is required.

- The order summary shows the complete payment breakdown: Applied Unused Voucher, New Voucher Issued (Unrestricted + Restricted), Wallet Payment, Revolving Fund Repayment (4%), and Total Wallet Charge.
- The reconciliation identity (Order = Total Voucher + Wallet Payment) is visible on the confirmation screen.

Agent: Restricted Phase Order

User Story: As a KoraLink Agent who has exceeded my Initial Unrestricted Ceiling, I want to understand the minimum wallet payment I need to make before a new voucher is issued, so that I can plan my cash flow.

Acceptance Criteria:

- The system displays the Minimum Required Repayment (70% of Net Outstanding Voucher) before showing the voucher allocation.
- The Agent sees a clear breakdown: Required Repayment amount, Additional Voucher to be issued, Remaining Wallet Payment, and Revolving Fund charge.
- If the Agent's wallet balance is insufficient to cover the Required Repayment + Revolving Fund charge, the order is blocked with a message indicating the minimum wallet balance needed.

Agent: Voluntary Repayment

User Story: As a KoraLink Agent, I want to make a voluntary repayment toward my Revolving Fund balance via the app, so that I can reduce my debt and potentially unlock higher voucher capacity.

Acceptance Criteria:

- The app/USSD menu includes a "Make Custom Repayment" option displaying the current Net Outstanding balance.
- The Agent can enter any amount from RWF 1 up to the total Net Outstanding.
- The system uses MVEND wallet integration to pull funds and apply them to the grant balance in real-time.
- The Progress Bar on the Agent dashboard updates instantly to reflect the new remaining balance.
- An SMS confirmation is sent: "RWF [Amount] received! Your new grant balance is RWF [New Balance]."
- Amounts exceeding Net Outstanding are rejected with a clear error message.

SFH Admin: Voucher Approval in Order Flow

User Story: As an SFH Admin, I want to review and approve order-embedded voucher requests so that I can control inventory distribution and manage fund risk within a single workflow.

Acceptance Criteria:

- The Admin Dashboard displays pending orders that include voucher requests, with the Agent's performance data (sales history, repayment rate, current Net Outstanding).
- Approving the voucher automatically approves the associated order and triggers stock reservation via the Stock Management Module.
- Rejecting the voucher automatically cancels the associated order and sends an SMS notification to the Agent with the rejection reason.
- The dashboard distinguishes between Unrestricted and Restricted phase vouchers for reporting purposes.

SFH Admin: Repayment Monitoring

User Story: As an SFH Admin, I want to view a comprehensive breakdown of Agent repayment activity including order-based deductions and voluntary payments, so that I can identify high-performing Agents and prioritize support for those falling behind.

Acceptance Criteria:

- The Admin Dashboard shows: Agent Name, Gross Voucher Received, Total Repaid, Net Outstanding, Unused Balance, Last Payment Date, and Payment Type (Order-Based / Voluntary).
- Admins can filter by repayment delinquency and sort by Net Outstanding to prioritize interventions.
- A "Send Reminder" bulk action is available for selected Agents.
- Audit trails link every repayment transaction to the original grant agreement for 10-year fiscal traceability.

Agent: Digital Service Float Order

User Story: As a KoraLink Agent ordering digital service float (refloating), I want the same voucher support available for float orders as for physical product orders so that I can start selling digital services without requiring upfront capital.

Acceptance Criteria:

- The voucher issuance engine applies the same Unrestricted/Restricted phase logic to float and refloating orders as it does to physical product orders.
- Before applying any voucher, the system checks in real time whether the digital service provider accepts voucher payment via their marketplace API integration. If the provider accepts voucher payment, the order proceeds through the standard Steps 1–7 voucher eligibility and issuance flow.
- If the digital service provider does not accept voucher payment for the order, the system blocks voucher allocation, displays a clear message to the Agent indicating that full wallet payment is required for this order, and proceeds with wallet payment only. The Revolving Fund repayment percentage is still applied to the wallet payment.
- The order summary and receipt display the payment breakdown in the same format as physical product orders: Applied Unused Voucher, New Voucher Issued (Unrestricted + Restricted where applicable), Wallet Payment, Revolving Fund Repayment, and Total Wallet Charge.
- The voucher amount issued for a digital service float order is deducted from the Agent's shared 360,000 RWF Maximum Allowed Voucher ceiling in the same way as physical product vouchers, and the reconciliation identity ($\text{Order Amount} = \text{Total Voucher Used} + \text{Wallet Payment}$) is enforced before the order is submitted.

System: DDIN Historical Float Data Migration

User Story: As the System, I must load historical voucher data for Agents who previously received float through DDIN so that their prior receipts are reflected in the Gross Cumulative Received Voucher balance from the point of their first marketplace transaction.

Acceptance Criteria:

- During Agent onboarding or platform data migration, the system accepts a pre-loaded historical voucher value per Agent representing the total float previously received through DDIN.
- This pre-loaded value is recorded as the opening Gross Cumulative Received Voucher balance for the Agent, before any marketplace transactions are processed.
- The system uses this opening balance immediately to determine the Agent's starting phase (Unrestricted or Restricted) and to enforce the correct repayment logic from the first order placed through the marketplace.
- Agents whose opening Gross Cumulative Received Voucher balance equals or exceeds the Initial Unrestricted Ceiling (50,000 RWF) are placed directly into the Restricted Phase from their first marketplace transaction.
- The pre-loaded historical balance is flagged in the audit trail as a migration entry, distinct from vouchers issued through the marketplace order flow, with the source recorded as DDIN and the migration date logged.
- SFH Admins can view each Agent's opening migration balance separately from marketplace-issued vouchers in the repayment monitoring dashboard.

Non-Functional Requirements

- **PIN-Based Security:** All wallet transactions, fund transfers, and voluntary repayments require PIN-based authentication.
- **Grant Agreement Linkage:** Every voucher issuance and repayment transaction must link back to the original legally enforceable grant agreement signed during Agent onboarding.
- **Audit Trail:** All voucher issuances, wallet payments, revolving fund deductions, and voluntary repayments must be logged in an immutable audit trail with timestamps, amounts, phase classification (Unrestricted/Restricted), and the associated order ID.
- **10-Year Retention:** All financial records generated by this module must be retained for the 10-year RRA fiscal retention period.
- **USSD Fallback:** Agents with basic phones must be able to make voluntary repayments via USSD shortcode with SMS confirmation.
- **Reconciliation Integrity:** The system must validate the reconciliation identity (Order = Total Voucher + Wallet Payment) on every transaction. Discrepancies must block the order and generate an alert for investigation.
- **Overpayment Prevention:** The Revolving Fund repayment must be capped at the Agent's remaining outstanding startup capital. The system must never collect more than what was disbursed.
- **Real-Time Dashboard Updates:** Agent progress bars, voucher capacity displays, and admin dashboards must reflect updated balances immediately upon transaction completion.

Stock Management & Inventory Module

Integration with Voucher System

Functional Requirements:

- **Real-time Stock Check:** The system must prevent agents from requesting a voucher for products that are currently out of stock at the SFH warehouse.
- **Automatic Balance Deduction:** Once SFH approves a voucher and the agent receives the goods, the voucher amount must be instantly debited from the agent's available credit limit and added to their Repayable Grant balance.
- **Proof of Delivery:** SFH must confirm the physical delivery of orders utilizing vouchers before the inventory is "active" in the agent's digital storefront.

Non-Functional Requirements

- **Auditability:** Every voucher request, approval timestamp, and subsequent stock movement must be logged in an immutable audit trail with unit-level granularity for 10-year fiscal traceability.
- **USSD Fallback:** Agents with basic phones must be able to request a voucher via USSD shortcode

6.6. Learning Management Module

This module delivers blended training, administers assessments, and manages the certification ecosystem, ensuring skills unlock new career flows.

Training & Certification

Purpose: To provide e-training and performance management tools, enabling KoraLink Agents to develop skills, earn certifications, and track progress.

Key Flow: Admin sets required, progressive learning tracks → Agent completes module (with offline download) → Agent passes product information test → System issues certificate/badge → Certifications populate profile.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to download modules for offline use during market days without the internet.

Acceptance Criteria: “Download” icon appears on video/PDF resources; content stores locally for 7 days.

User Story: As a KoraLink Agent, I want my upcoming training sessions to auto-populate my calendar so I don’t miss them.

Acceptance Criteria: Pulls from backend eLearning module; SMS reminder 1 hour before.

Assessment & Career Integration

Purpose: To verify proficiency and integrate learning outcomes directly into the employment marketplace.

Key Flow: Agent fails quiz → System provides “Review Mistakes” button → Agent retakes quiz (max 3 attempts) → Certification is issued, unlocking eligibility for jobs.

User Stories & Acceptance Criteria:

User Story: As the System, I want to auto-revoke expired certifications (e.g., after product updates) and prompt retraining.

Acceptance Criteria: Blocks related gigs until renewal; sends SMS notification.

Non-Functional Requirements

- Content must support multi-format content and translations (Kinyarwanda/English).
- Training modules must provide practical engagement or “sandbox” environments for Agents to place real orders.

Social Learning & Learner Feedback

Purpose: To foster peer-to-peer engagement and provide a direct communication channel between facilitators and KoraLink Agents for clarifying key concepts and improving training content.

Key Flow: Facilitator/Agent creates a topic → Notification sent to cohort → Agents participate in threaded discussion → Learners submit module feedback.

User Stories & Acceptance Criteria:

- Facilitator-Led Discussions

User Story: As a Course Facilitator, I want to trigger discussion threads on key concepts within a module so that I can clarify complex topics and engage with Agents in real-time.

Acceptance Criteria:

- Facilitators can pin "Concept Deep-Dive" threads to specific training modules.
- Supports rich text and links to related module resources.
- Facilitators receive alerts for new replies via the Notification System.

• Agent Participation & Peer Learning

User Story: As a KoraLink Agent, I want to participate in forum discussions with my peers and facilitators so that I can ask questions and share practical insights from the field.

Acceptance Criteria:

- Agents can reply to threads using the mobile app.
- Offline Support: Agents can draft forum replies offline, which auto-sync when connectivity is restored.
- Ability to "Like" or "Upvote" helpful peer responses.

• Learner Feedback Loop

User Story: As a Learner (KoraLink Agent), I want to provide feedback on the clarity and usefulness of a module so that SFH can improve future training materials.

Acceptance Criteria:

- A "Feedback" button appears at the end of every completed module.
- Supports both structured (1–5 star rating) and unstructured (text comments) feedback.
- Privacy: Option to submit feedback anonymously to the SFH Admin.

Functional Requirements

- Threaded Discussions: The system must support hierarchical comment structures for organized conversations.
- Facilitator Dashboard: SFH Admins must have a view to moderate discussions and see aggregated learner feedback scores.
- Notification Integration: New forum topics or direct replies must trigger SMS/Push notifications based on the user's preference.

Non-Functional Requirements

- Data Minimization: Forum data should be stored efficiently to ensure the Next.js/PostgreSQL backend remains performant during high-traffic discussion periods.

- **Moderation Logs:** The system must maintain audit logs for all deleted or flagged forum content to ensure compliance and traceability

6.7 Agent Performance-Based Incentive Module

This module automates the calculation of monthly Agent performance scores and manages the issuance of quarterly incentive vouchers. It operates as a policy-driven scoring service that consumes operational data from the KoraLink Marketplace and the external phone repayment management system (IntelliGra), without owning the underlying transactions.

Monthly Scoring & Tier Assignment

Purpose: To calculate each KoraLink Agent's monthly performance score across three operational dimensions and assign a corresponding performance tier that reflects their activity, repayment discipline, and business growth.

Key Flow: System loads active incentive policy configuration → Checks whether Agent is in a grace period (excluded from scoring) → Requests phone repayment data from IntelliGra → Calculates Repayment score → Calculates Sales score → Calculates Orders score → Computes total monthly score → Assigns performance tier → Persists scoring record → Generates audit log.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to see my current monthly score broken down by component so that I can understand what is driving my performance and what I need to improve.

Acceptance Criteria: The Agent's dashboard displays three separate score cards Repayments, Sales, and Orders each showing the points earned versus the maximum possible, the input values used (e.g., repayment percentage, total sales value in RWF, total paid order value in RWF), and whether any minimum threshold was missed. The dashboard also displays the Agent's current tier (Bronze / Silver / Gold / Platinum) and total points for the month.

User Story: As a KoraLink Agent, I want to see my quarter-to-date status and whether I remain on track for quarterly voucher eligibility so that I am not surprised at the end of the quarter.

Acceptance Criteria: A quarter-to-date summary is visible on the Agent's dashboard showing: cumulative monthly scores for the current quarter, current quarterly phone repayment compliance

percentage against the 80% threshold, and a plain-language eligibility status indicator (e.g., “On track” or “At risk — repayment below threshold”).

User Story: As an SFH Admin, I want to view a summary of all Agent tier assignments for the current month so that I can monitor program performance and identify Agents who need support.

Acceptance Criteria: The Admin dashboard displays a filterable table of Agents with their monthly tier, total points, and component breakdown. Admins can filter by tier, district, and cohort.

Repayment Scoring

Purpose: To reward Agents for maintaining consistent repayment of their phone installment obligations, sourced from the IntelliGra external repayment system.

Key Flow: System requests monthly repayment data from IntelliGra → Computes repayment percentage ($\text{actual repayment} \div \text{expected repayment}$) → Maps percentage to configured points threshold → Awards repayment score.

Note: Agents who did not receive a smartphone through the KoraLink program are excluded from phone repayment scoring. Starter stock recovery scoring is defined under the Orders dimension (reinvestment rule, Section 6.X below). Future activation of product/service startup capital repayment scoring is noted as a planned extensibility point.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent who received a smartphone through the program, I want to see my expected monthly repayment amount and what I have paid so that I can verify my repayment score is correct.

Acceptance Criteria: The Agent’s repayment score card displays: expected monthly repayment ($392 \text{ RWF} \times \text{days in month}$), actual repayment received per IntelliGra, calculated repayment percentage, and the points awarded based on the active threshold table. If the Agent has no phone obligation, the card is hidden and repayment points are excluded from their scoring.

User Story: As the System, I must apply the active phone repayment scoring thresholds as configured, without hardcoding values into the application logic.

Acceptance Criteria: The scoring thresholds table (repayment percentage → points) is stored in the policy configuration layer and is applied dynamically at calculation time. Changes to thresholds take effect on the next scheduled scoring run without requiring a code deployment.

The current active thresholds are:

Repayment Percentage	Points Awarded
≥ 100%	20
≥ 90%	15
≥ 80%	10
≥ 70%	5
< 70%	0

Sales Scoring

Purpose: To reward Agents for consistent selling activity, measured through inventory reduction recorded in the platform rather than individual customer transactions.

Key Flow: Agent records sales in the platform → Inventory auto-deducts → System aggregates total monthly sales value using RRP as the default (or Agent-entered higher price where applicable) → Maps total value to configured points threshold → Awards sales score.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to see the total sales value the platform used to calculate my score so that I can verify it matches what I recorded.

Acceptance Criteria: The sales score card displays total monthly sales value in RWF, the price basis used (RRP or Agent-entered higher price per transaction), whether the 50,000 RWF minimum threshold was met, and the points awarded. If the threshold was not met, a clear message states: “Sales below 50,000 RWF — 0 points awarded this month.”

User Story: As the System, I must apply the active sales scoring thresholds as configured, without hardcoding values into the application logic.

Acceptance Criteria: The sales thresholds table is stored in the policy configuration layer and applied dynamically. The current active thresholds are:

Monthly Sales Value (RWF)	Points Awarded
Below 50,000	0
50,000 – 100,000	10
100,001 – 150,000	20
150,001 – 250,000	30
250,001 – 350,000	40
Above 350,000	50

Orders Scoring

Purpose: To reward Agents for responsible restocking behaviour, including reinvestment of their remaining unpaid starter stock grant balance, ensuring the Revolving Fund is used for its intended purpose.

Key Flow: System aggregates total paid and confirmed orders for the month → Checks minimum conditions (value $\geq$ 50,000 RWF and frequency $\geq$ 2 orders) → Checks reinvestment rate against remaining unpaid starter stock grant balance → Maps order value to configured points threshold → Applies reinvestment cap if applicable → Awards orders score.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to see my total paid order value and whether my reinvestment rate met the 70% rule so that I understand whether my Orders points were capped.

Acceptance Criteria: The orders score card displays: total paid order value in RWF for the month, number of confirmed paid orders, whether both minimum conditions were met (value $\geq$ 50,000 RWF and count $\geq$ 2), the reinvestment rate percentage against the remaining unpaid starter stock grant balance, whether the 70% reinvestment rule was met, and the points awarded. If the reinvestment rule was not met, the card shows: “Orders points capped at 20 — reinvestment below 70%.”

User Story: As the System, I must apply the active orders scoring thresholds and reinvestment cap as configured, without hardcoding values into the application logic.

Acceptance Criteria: The orders thresholds table and reinvestment cap are stored in the policy configuration layer. The current active thresholds are:

Monthly Total Paid Order Value (RWF)	Points Awarded
Below 50,000	0
50,000 – 100,000	10
100,001 – 200,000	20
200,001 – 300,000	25
Above 300,000	30

If the Agent’s reinvestment rate is below 70% of their remaining unpaid starter stock grant balance, Orders points are capped at 20 regardless of order value.

Quarterly Eligibility Validation & Voucher Issuance

Purpose: To determine, at the end of each quarter, which Agents qualify for incentive voucher rewards based on their monthly scores and sustained repayment discipline across the quarter.

Key Flow: Quarter ends → System aggregates three monthly scores → Checks all eligibility gates → Assigns final quarterly tier → Issues voucher to eligible Agents → Records issuance in audit log → Notifies Agents via SMS.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to receive an SMS notification at the end of the quarter confirming my tier and voucher amount so that I know the outcome without having to log into the app.

Acceptance Criteria: Upon quarterly processing, the system sends an SMS to every Agent with a tier result. For eligible Agents the SMS states their tier, points, and voucher amount (e.g., “Congratulations! You are a Gold Agent this quarter 23,000 RWF voucher issued.”). For ineligible Agents the SMS states their tier and the reason for ineligibility (e.g., “Your quarterly repayment compliance was below 80% no voucher issued this quarter.”).

User Story: As an SFH Admin, I want to see a quarterly eligibility report showing which Agents qualified for vouchers, at which tier, and the total voucher value to be disbursed so that I can prepare for distribution.

Acceptance Criteria: The Admin dashboard generates a quarterly summary report filterable by tier, district, and cohort. The report includes each Agent’s three-month score history, final tier, eligibility status with reason (if ineligible), and the total voucher liability for the quarter.

User Story: As the System, I must enforce all eligibility gates before issuing any voucher reward.

Acceptance Criteria: An Agent only receives a voucher if ALL of the following conditions are met: (1) total monthly points ≥ 50 in the relevant month, (2) quarterly phone repayment compliance $\geq 80\%$ across the three-month quarter, and (3) no confirmed major compliance violations or fraud findings exist for the period. Failure of any single gate results in no voucher for that quarter, regardless of points achieved.

The current active tier and voucher configuration is:

Tier	Points Range	Quarterly Voucher
Platinum	85 – 100	30,000 RWF
Gold	70 – 84	23,000 RWF
Silver	50 – 69	15,000 RWF
Bronze	Below 50	No voucher

Sanctions & Compliance

Purpose: To protect the integrity of the incentive system by recording and applying point deductions and eligibility blocks when Agents breach program rules.

Key Flow: Compliance officer or system flags violation → Sanction level determined → Points deducted or eligibility blocked → Sanction recorded with reason in audit log → Agent notified.

User Stories & Acceptance Criteria:

User Story: As a KoraLink Agent, I want to be able to see any sanction applied to my score and the reason for it so that I can understand and, where applicable, resolve the issue.

Acceptance Criteria: The Agent's dashboard displays a sanctions history log showing: date applied, sanction level, reason, and impact on points. Sanctions are visible but not editable by the Agent.

User Story: As an SFH Admin, I want to apply a sanction to an Agent's score with a mandatory reason field so that every deduction is traceable.

Acceptance Criteria: The Admin interface requires a reason text entry before a sanction can be saved. The sanction is immediately reflected in the Agent's score and logged in the immutable audit trail.

The current active sanctions framework is:

Sanction Level	System Action
Minor non-compliance	–5 points
Moderate non-compliance	–15 points and voucher held pending resolution
Major non-compliance / fraud	Disqualification from quarterly voucher and escalation under programme rules

Non-Functional Requirements

- The incentive system must operate as a policy-driven scoring service. All scoring thresholds, tier definitions, voucher values, eligibility rules, and sanction levels must be stored in a configurable policy layer and must not be hardcoded into application logic. Changes to policy must take effect on the next scheduled scoring run without requiring a code deployment.
- Monthly scoring must execute on a scheduled basis, typically on the first day of the following month, and must be capable of processing all active Agents without manual intervention.
- The system must maintain a complete and immutable audit trail for every scoring event, covering: repayment evaluation responses received from IntelliGra, all component score calculations and the policy version applied, eligibility decisions and the gate conditions evaluated, voucher issuance records, sanction applications with reasons, manual overrides with authoriser identity, and recalculation history. All prior scoring outcomes must remain reconstructable from source data and the historical policy configuration, even after future policy changes.
- Agent scoring data must be accessible to the Agent only for their own record. SFH Admins have read access to all Agent records within their assigned scope. Unauthorised cross-Agent data access must be blocked at the access control layer.
- The module must integrate with IntelliGra via API for phone repayment data retrieval. The integration must handle IntelliGra unavailability gracefully if data cannot be retrieved within the scoring window, the system must flag the affected Agent's repayment score as pending, notify the Admin, and allow for a manual trigger of the repayment scoring step once data is available.
- Scoring records and audit logs must be retained for a minimum of 10 years in line with the RRA fiscal retention requirement.
- The architecture must support future extensibility including: activation of product and service startup capital repayment as an additional scoring dimension, cohort-specific scoring policy configurations, and integration of fraud detection rules as an automated sanction trigger.

6.8 Job Management Module

This module formalizes the training-to-employment pipeline, connecting upskilled Agents with verified Employers based on trackable performance and certified skills.

Purpose: To formalize the training-to-employment pipeline by connecting upskilled KoraLink Agents with verified Employers through a closed-loop ecosystem.

Employer Experience (Web Portal)

Key Goal: To provide recruiters and SMEs with a streamlined "Hiring Pipeline" to manage talent at scale.

- **Employer Dashboard:** A real-time operational snapshot showing active postings, weekly applicant volume, and a 14-day trend chart to gauge engagement.
- **Job Creation with Live Preview:** A multi-section form (Job Basics, Compensation, Description, Responsibilities) that includes a right-side "Live Preview" panel, allowing employers to see exactly what the posting will look like on the Agent's mobile app before publishing.
- **Applicant Evaluation & "Fit Scoring":**
 - **Fit Score:** A visual bar and percentage (e.g., Green > 80%) indicating how well an Agent matches the job requirements based on their certifications and sales history.
 - **Applicant Detail View:** Full access to Agent profiles, including cover letters, skills chips, and internal recruiter notes for private team evaluation.

Pipeline Management: Centralized status tracking to move candidates through stages: **New** → **Shortlisted** → **Interviewed** → **Hired/Rejected**.

Agent Experience (Mobile App)

Key Goal: To empower KoraLink Agents to browse, save, and apply for work opportunities that match their certified skills.

- **Job Marketplace (Opportunities):** A dedicated tab for Agents to browse listings by location, pay rate, and duration.
- **Saved Opportunities:** Agents can bookmark jobs to apply later, with saved items persisting across sessions.
- **One-Tap Application Flow:**
 - **Auto-Population:** The application form is pre-filled with the Agent's profile data, education, and SFH-verified certifications.
 - **Attachments:** Agents can upload ID cards, health certificates, or motivation letters (PDF/JPG) directly from their mobile device.
- **Application Tracking & Feedback:**

- My Applications: A view showing the status of all current applications (Accepted, Rejected, or New).
- Employer Feedback: Agents receive direct inline feedback (e.g., "Congratulations! We have selected you...") and a history timeline of their application status.

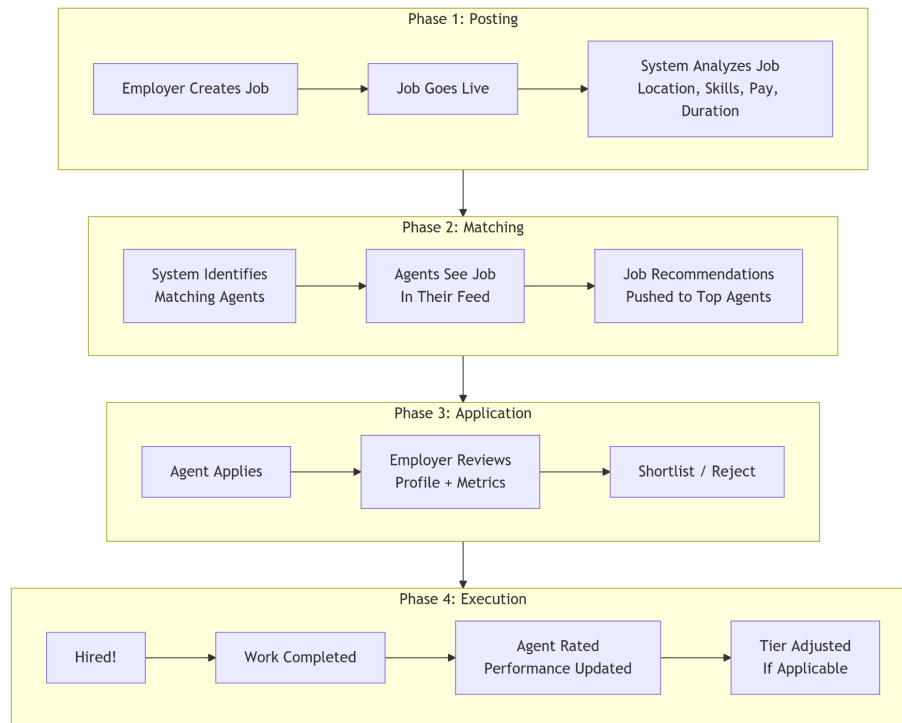
Performance & Reputation (The Feedback Loop)

Purpose: To build trust and long-term career ladders.

- Post-Gig Ratings: After an Agent completes a job, the Employer submits a rating and review which automatically updates the Agent's "Reputation Metrics" on their public profile.
- Asymmetric Data Visibility: To maintain professional standards, Fit Scores are visible only to the Employer, while Employer Feedback and Status Updates are visible to the Agent.

Functional & Non-Functional Additions

- Messaging Module: A communication channel for Employers to reach out to shortlisted talent.
- Notification Integration: Real-time alerts for both sides:
 - Employer: Notified of new applicants and expiring posts.
 - Agent: Push notifications for shortlisting, rejections, and when an employer views their application.
- Auditability: The system must maintain immutable audit logs for all hiring actions, interview outcomes, and job status changes.



The flow illustrated in the image represents the End-to-End Job Lifecycle within the KoraLink ecosystem, detailing how the platform's backend intelligence connects employers with upskilled agents. This flow bridges the Employer Web Portal and the Agent Mobile App into a single, closed-loop process.

Phase 1: Posting

The process begins when an Employer utilizes the web portal to create a job opportunity.

- **Template-Driven Creation:** Employers can use pre-approved templates to ensure compliance and consistency.
- **Real-time Analysis:** As the job is drafted, the system utilizes a "Live Preview" so the employer sees exactly what the agent will see on mobile. Simultaneously, the system analyzes the job's location, required skills, pay, and duration to prepare for matching.

Phase 2: Matching (The Matchmaking Engine)

Once the job goes live, the platform's backend (Next.js/PostgreSQL) acts as a digital matchmaker.

- **System Identification:** The system scans the KoraLink Agent Registry to identify agents whose certified skills and verified location match the job's requirements.
- **Intelligent Feed:** Agents see these matching jobs in their "Opportunities" tab.
- **Push Recommendations:** For high-priority roles, the system "pushes" recommendations to the top-performing agents (those with high ratings and certifications) to ensure the employer receives quality applicants.

Phase 3: Application

This phase moves the interaction from discovery to formal engagement.

- **One-Tap Application:** Agents can apply instantly; the system auto-fills their application with SFH-verified data, certifications, and sales history from their profile.
- **Asymmetric Fit Scoring:** The employer reviews applicants via a dedicated dashboard that features a "Fit Score" (a visual bar/percentage matching agent skills to job needs). This score is intentionally hidden from the agent to maintain professional screening standards.
- **Pipeline Management:** The employer then moves the candidate through the status stages: Shortlisted or Rejected.

Phase 4: Execution

The final phase focuses on the actual work and the data-driven results that follow.

- **Verification of Work:** Once "Hired," the agent completes the work. The platform tracks this as part of the agent's professional history.
- **Reputation Update:** Post-gig, the employer provides a rating and performance review. These ratings are immutable and update the agent's public-facing "Reputation Metrics".
- **Career Progression (Tier Adjustment):** If the performance is high, the system may trigger a Tier Adjustment. This supports the KoraLink progression model, helping agents move from Level A (survival income) toward Level F (small business employer) based on their cumulative performance data.

7. Non-Functional Requirements

Non-Functional Requirements define the essential quality attributes necessary for the Koralink Digital Platform to operate securely, reliably, and compliantly within the constraints of the Rwandan regulatory and operational environment. These NFRs ensure the system can effectively support the program's scale, targeting over 11,762 Koralink Agents onboarded across four years.

Technical Architecture and Scalability

The platform utilizes a modern, decoupled web stack designed for performance and scale, specifically excluding dependence on a legacy ERP base.

- **Technical Stack Adherence:** The system must utilize a Next.js API backend and a PostgreSQL database (leveraging Prisma ORM). The mobile application must be built with React Native + Expo,.
- **Scalability:** The architecture must support significant growth in user volume and handle high transaction loads. The design must allow for horizontal scaling of background processes and notification handling.
- **Load Handling:** The system is required to handle expected transaction loads of 5–20 transactions per second during regular use, with the capacity to manage bursts up to 50–100 per second.
- **Integration Standards:** The platform requires standardized integration with external services, utilizing secure APIs and webhooks. Critical integrations include Twilio for SMS services and SendGrid for email delivery.

Security and Privacy Compliance

Strict measures must be implemented to protect the high volume of Personally Identifiable Information (PII) and financial data, ensuring compliance with Rwandan law and financial security standards.

- **Legal Compliance:** The Organization must comply with Rwanda's Protection of Personal Information Law,. This mandates completing registration as a Data Controller with the NCSA (National Cyber Security Agency).
- **Data Security:** All Personal Data must be encrypted at rest and in motion. The platform must use HTTPS-only communication.
- **Consent Management: Mandatory Consent Checkboxes** for PII processing must be implemented during all registration flows. The system must also support the data subject's right to erasure.
- **Authentication and Access Control:** Authentication must utilize JWT token-based authentication. The system requires field-level permissions and encrypted PIN storage for sensitive operations.
- **Financial Security:** The Wallet module requires PIN-based security for transactions and withdrawals. Card payment processing must ensure PCI-DSS Level 4 certification compliance by tokenizing card numbers.
- **Incident Response:** A defined Data Breach Incident Response Plan must mandate external reporting of any Personal Data breach to the NCSA within 48 hours.

Compliance and Data Retention

The platform must satisfy mandated fiscal and contractual compliance requirements, prioritizing RRA mandates and ensuring full auditability.

- **RRA Tax Compliance:** Mandatory adherence to the RRA EBM E-Invoice System is required. The system must obtain official RRA EBM certification before going live.
- **Invoice Automation:** The system must implement auto-submission of invoices to RRA for all customer invoices,, and mandate TIN Validation for Employers.
- **Offline Protocol:** A VSDC fallback mechanism (protocol) must be implemented to log and retry failed invoice submissions immediately when connectivity is restored, ensuring all transactions remain fiscally compliant.

- **Data Retention Policy:** The platform must adopt the 10-year RRA requirement for fiscal and financial records, as this rule overrides the Foundation's 7-year minimum retention period.
- **Auditability:** An immutable audit trail must be maintained across all financial, inventory, and administrative actions to satisfy funder and regulatory audit requirements,. Inventory movements must be logged with unit-level granularity.
- **Legal Agreements:** The system must mandate digital signing and secure storage of the legally enforceable grant agreement (Repayable Grant terms) during Agent onboarding for Revolving Fund traceability.

Performance, Reliability, and Accessibility

The system must guarantee high reliability for continuous operations and provide accessibility for Agents operating in rural, low-connectivity contexts.

- **Performance:** Initiating a payment request must take less than one second under normal conditions.
- **Reliability:** All payment-related services must maintain at least 99.9% uptime to ensure continuous business operations for Koralink Agents. Downtime should be minimized.
- **Accessibility:** The platform must maintain integrated offline USSD/SMS fallback for low-connectivity contexts,. Koralink Agents must be able to record sales and check earnings/fund status via USSD,.
- **Usability and Localization:** The platform must provide a user-friendly interface for all stakeholders, with intuitive navigation. The system must support both Kinyarwanda and English languages for training content and user interfaces.
- **Transaction Integrity:** The system must ensure that duplicate or conflicting payment requests are avoided to maintain data integrity.

8. Constraints and Assumptions

This section outlines the non-negotiable limitations imposed upon the development and deployment of the Koralink Digital Platform, alongside the operational and technical assumptions necessary for the system to achieve its intended objectives within the Rwandan market.

Platform Constraints

The development and deployment of the platform are bound by financial, temporal, and technical limits that must be strictly adhered to.

The project timeline is governed by the specific Agreement Term, which extends from March 1, 2025, to August 31, 2029, and development must adhere to the specified budget allocated within the overall Fund. Furthermore, the platform development must adhere to the evaluation timeline set by the Mastercard Foundation in Year 3 for potential expansion.

The system architecture is strictly mandated to utilize a modern decoupled stack: a Next.js API backend and a PostgreSQL database (Prisma ORM). The mobile application must be built using React Native + Expo.

Security standards are non-negotiable, requiring that authentication be JWT token-based, and all sensitive data must be encrypted. Operationally, the platform must maintain integrated offline registration/sales capabilities via USSD/SMS fallback to support users in low-connectivity contexts. Additionally, the platform must include mechanisms to track Repayable Grants and their terms to ensure the integrity of the Revolving Fund.

Operating Assumptions

Successful implementation relies on the validity of several key external factors, partner commitments, and regulatory conditions.

The program assumes that the target youth population in Rwanda has sufficient access to the provided smartphones or devices and adequate internet connectivity and data necessary to utilize the platform effectively.

Partner cooperation is essential, assuming that key implementing partners, including SFH, will provide central and regional warehouses for distribution of goods, and the ICT Chamber will provide platform customization, deployment, operation, and maintenance services.

For financial operations, the successful integration of external payment gateways (such as Gwiza Payment Gateway and DDIN Digital Services) and digital service APIs is assumed. Furthermore, it is assumed that the licensed third-party wallet provider maintains necessary RURA licensing and security standards (such as PCI-DSS).

The regulatory environment in Rwanda is assumed to remain supportive of digital commerce and the iWorker model. Crucially, the system relies on securing Data Agreements with key agent networks (MOMO, MTN, Equity) to enable the digital income history validation required for the recruitment process. Finally, it is assumed that KoraLink Agents are willing and able to make repayments toward the Revolving Fund.

9. Dependencies

Successful implementation and sustained operation of the Koralink Digital Platform relies on the timely delivery of key outputs and continued cooperation from external partners and regulatory bodies. These dependencies are critical, as their failure would severely impact the platform's ability to achieve its strategic objectives, particularly Agent onboarding and financial traceability.

Operational and Partner Dependencies

The platform's success is fundamentally reliant on the execution of non-digital functions managed by key implementing partners.

- **SFH Operational Input:** The execution of the Agent lifecycle relies on SFH operational input, including effective Agent recruitment, training, and ongoing management,. SFH is

also responsible for managing the physical distribution and logistics of health goods from central and regional warehouses to Koralink Agents.

- **ICT Chamber Services:** The technical continuity of the platform depends on the ICT Chamber providing the necessary platform customization, deployment, ongoing operation, and maintenance services.
- **Phone Repayment Data (IntelliGra):** The Agent Performance-Based Incentive Module depends on timely data retrieval from IntelliGra, the external phone repayment management system. Failure to retrieve repayment data within the monthly scoring window will result in repayment scores being held as pending until data is available. A manual trigger mechanism must be available to complete scoring once data is restored.
- **Digital Service Provider APIs:** The voucher issuance flow for digital services (float and refloating) depends on digital service providers maintaining active API integrations with the marketplace platform. Providers must maintain open wallets to enable Agent wallet payments for refloating. Additionally, DDIN historical float data must be successfully loaded into the marketplace as a one-time migration before the digital services voucher module can go live. This migration is a hard prerequisite for accurate Net Outstanding Voucher calculations for Agents who previously received float through DDIN.
- **External Partnerships:** The expansion of the marketplace requires the establishment and maintenance of partnerships with health goods/services providers, telecommunication companies, and device manufacturers. Specific services like TeleHealth (Irembo) and Health Insurance (EdenCare) are being explored as potential integrations.
- **Infrastructure Adoption:** Successful adoption of related program components, such as the physical training sessions and the network of eBusiness hubs/digital service centers, is necessary to support the Agent network beyond the digital interface.

Financial and Legal Dependencies

Financial stability and the legal enforceability of the Revolving Fund are non-negotiable prerequisites for the program's continuity.

- **Program Funding:** The continuation of Services is dependent on satisfactory progress and delivery as determined by the Foundation, which manages the overall Program budget (approximately US \$25,064,916). This includes the timely provision of Fund disbursements from the Foundation (or Fund Manager, if retained) to the Organization.
- **Revolving Fund Agreements:** The provision of Repayable Grants (in the form of goods, devices, accessories, and data) requires a written, legally enforceable agreement with the Koralink Agent detailing repayment terms, which the platform must digitally sign and secure.
- **Deliverable Acceptance:** The continuation of the project relies on Client (Foundation/Organization) review and acceptance of platform deliverables, such as the completion of User Acceptance Testing (UAT).

Regulatory and Data Dependencies

Platform integrity requires alignment with national regulations, particularly those governing finance and data access.

- **Regulatory Alignment:** Policy engagement with the RRA (Rwanda Revenue Authority), the Ministry of Health, and the FDA is required to secure clear operating guidelines and tax treatment. Crucially, the platform must obtain official RRA EBM certification before the sales module can go live.
- **Financial Licensing:** The success of the financial management module assumes that the licensed third-party wallet provider maintains necessary RURA licensing and security standards (such as PCI-DSS).
- **Data Agreements for Recruitment:** The execution of the recruitment process, which validates digital income history, is dependent on securing Data Agreements with key agent networks (MOMO, MTN, Equity) for income verification.

10. Success Criteria and KPIs



The success of the Koralink Digital Platform is measured by its capacity to achieve the strategic program objectives of poverty relief and digital jobs creation, as defined by the Agreement with the Mastercard Foundation. Success criteria are defined by quantitative targets and qualitative assessments that track Agent progress, financial integrity, and operational efficiency.

Economic Empowerment and Livelihoods

The primary success of the platform is tied to the measurable financial outcomes of the KoraLink Agents and the overall job creation impact.

The ultimate quantitative success metric is the average monthly earning of KoraLink Agents via the Marketplace Platform, with an escalating target that reaches \$310 per month by the final year of the program. This target is rooted in earlier phases, aiming for \$100 per month in Year 1. Total job creation must meet the goal of facilitating 14,640 direct and 29,360 indirect work opportunities. Success is also measured by the platform's effectiveness in enabling Agents to increase earnings and access sustainable livelihoods. Furthermore, livelihood improvement is tracked by assessing the utilization and offering of incentives such as health insurance.

The financial integrity of the Revolving Fund is a key success area, measured by the utilization rate of the Fund and the repayment rates of the Repayable Grants. The platform must actively track automatic deduction rates and hardship case approvals related to fund repayment.

[Pending policy input KPIs for the Agent Performance-Based Incentive Module, including quarterly tier distribution targets and voucher disbursement rates, to be defined by TCP Africa and SFH in the next BRD update cycle.]

Agent Lifecycle and Operational Targets

Success is dependent on the platform's ability to onboard and progress Agents effectively while adhering to demographic commitments.

The platform must facilitate the onboarding of a total target of 11,762 Agents across four years. The program must also meet specific demographic targets, ensuring that 80% to 90% of Agents are women and 70% are rural-based.

Success in skills development is tracked through completion rates for e-training modules and the number of certifications earned. Certifications and badges must signify progression across learning levels, which is critical for supporting the intended career progression model (from Level A to Level F). Job marketplace success is measured by the number of employers registered and the jobs posted and filled via the platform. Additionally, tracking whether Agents who drop out secure jobs outside the platform is relevant to assessing the program's full impact.

Operational efficiency is measured by metrics such as onboarding speed (time from application to approval) and overall order/case cycle time.

Platform Quality and Traceability

The quality attributes of the platform ensure trust and compliance, which are essential for long-term sustainability.

- **Reliability and Performance:** The platform must maintain high service availability, particularly for payment-related services, with a minimum 99.9% uptime. Transaction performance is key, requiring that the initiation of a payment request takes less than one second under normal conditions. The platform must ensure high payment success rates (at least 95% within one minute) and strict transaction integrity, avoiding duplicate or conflicting requests (target conflict rate of less than 0.05%),.
- **Compliance and Audit:** Compliance with privacy and security requirements, including Rwanda's Protection of Personal Information Law, is mandatory. Success is achieved by delivering full traceability through an immutable audit trail for every transactional and system event. Furthermore, reconciliation accuracy should be extremely high, aiming for less than 0.1% of transactions remaining unmatched after reconciliation.

- User Satisfaction: Qualitative success is measured by user satisfaction across all stakeholder groups KoraLink Agents, Consumers, and Employers based on feedback and platform usage data.

Reporting and Governance

The system must satisfy mandated reporting requirements to allow the Foundation to evaluate the program's success. This requires the platform to provide role-based dashboards (Admin, Agent, Employer, Analytics) and the capability to generate exportable, API-accessible financial, sales, job, and M&E reports. Success is achieved when these detailed reports are available for fulfilling quarterly narrative and financial reports, as well as ad-hoc requests.

11. Reporting Requirements

The Koralink Digital Platform must provide comprehensive reporting capabilities that ensure transparency, regulatory compliance, and robust data insights necessary for the Foundation and operational partners to govern the program and measure success against key performance indicators (KPIs).

Funder Mandates and Financial Governance

Reporting mandates are dictated by the Foundation's Agreement, necessitating detailed, scheduled, and on-demand data delivery.

The system must support the generation of mandatory Quarterly narrative and financial reports detailing performance against KPIs, starting with the quarter ending June 30, 2025. The platform is also required to support Annual Narrative Reports and Detailed Expenditure Statements. The platform should facilitate access to necessary digital records for inspection/evaluation by Foundation representatives.

All financial, sales, job, and M&E reports must be both exportable and API-accessible for easy review by funders and regulators. The Fund Manager (if retained) requires access to financial data and specific tools to generate reports related to repayable grants and commission investigations. The system must also retain financial records for a minimum of ten years to comply with RRA mandates, overriding the Foundation's seven-year requirement for fiscal data.

Operational and Compliance Monitoring

The system must provide the tools necessary for operational supervisors and compliance officers to manage agents and adhere to national fiscal laws.

Agent Performance and Livelihoods

The system must generate performance reports necessary for evaluating program goals, tracking metrics such as Agent onboarding, total earnings, job placements, and training completion rates. Supervisors require specific reports to track long-term program outcomes, including whether Agents who drop out secure jobs outside the platform, to assess the program's full impact.

Revolving Fund Traceability

High-level reporting is required for the Revolving Fund, including metrics on Grant Utilization and Recovery Rates. Dedicated reports must allow SFH Admins to filter Agents by repayment delinquency and overdue amounts to prioritize support interventions.

Regulatory and Audit Logs

Compliance is ensured through mandatory record-keeping. The platform must maintain a comprehensive access log, device/session fingerprinting, and an immutable audit trail for every transactional and system event. For RRA tax compliance, the system must continuously track EBM submission logs and exception reports, alerting Admins if the VSDC fallback is triggered due to connectivity issues. Inventory control requires reports on Agent Stock, Sales by Location, and an audit trail for all product movements logged with unit-level granularity.

Incentive Scoring and Voucher Disbursement Reporting

The platform must generate monthly incentive scoring reports showing each Agent's component scores (Repayments, Sales, Orders), total points, tier assignment, and eligibility status for the period. Quarterly reports must summarise voucher disbursement by tier, total RWF value disbursed, and flag any Agents who failed eligibility gates with the reason recorded. Sanctions logs must be exportable and retained as part of the immutable audit trail for the 10-year fiscal retention period.

Data Accessibility and Analytics

Accessibility is provided through customizable dashboards and data export capabilities for various stakeholders.

- **Dashboards:** Role-based dashboards must be provided for all major user types, including Admin, Agent, Employer, and Analytics roles. These dashboards track critical data relevant to the user's role, such as personal sales, stock levels, commission earnings, and application overviews.
- **Data Export:** The platform must provide the capability to generate Ad-hoc Reports upon written request from the Foundation. All financial, sales, job, and M&E reports must be exportable and API-accessible to satisfy external governance requirements.
- **Data Integrity:** A data governance framework and validation rules must be implemented to ensure the accuracy and reliability of all information used for reporting and value-added services.

12. Data Management and Privacy

This section defines the mandatory requirements for data governance, security, retention, and privacy compliance, ensuring that the platform processes the Personal Data (PII) of Koralink

Agents and consumers securely and in adherence to Rwandan law and contractual obligations. The Organization acts as an independent data controller in this process.

Data Privacy and Security Compliance

The platform must embed strict technical and administrative controls to protect sensitive data and meet regulatory mandates.

- **Data Protection Law Adherence:** Compliance with all applicable privacy laws is mandatory, including Rwanda's Protection of Personal Information Law (Law N°058/2021).
- **Data Controller Registration:** The Organization must complete the mandatory registration as a Data Controller with the NCSA (National Cyber Security Agency).
- **DPO Formalization:** A Data Protection Officer (DPO) must be formally assigned to oversee PII oversight and breach notification management.
- **Consent Management:** The platform must obtain all necessary informed consents and/or provide transparent disclosures required by law, mandating explicit Consent Checkboxes for PII processing during all registration flows (Agent and Consumer).
- **Data Subject Rights:** The system must incorporate features to support data subject access protocols, including the right to erasure.
- **Data Security:** All Personal Data must be encrypted at rest and in motion, and security measures must include API security (HTTPS-only) and encrypted PIN storage.
- **Incident Response:** A defined Data Breach Incident Response Plan must be developed and implemented, requiring mandatory external reporting of any Personal Data breach to the NCSA within 48 hours.

Data Retention and Governance

The system must ensure high data quality and adhere to the strictest retention policies dictated by Rwandan fiscal law.

- **Dual Data Retention Policy:** The platform must adopt the 10-year RRA requirement for fiscal and financial records, such as sales, commissions, and repayment logs. This duration overrides the Foundation's 7-year minimum retention period for digital records maintained post-Agreement termination, ensuring compliance and audit readiness for tax purposes.
- **Data Governance Framework:** A data governance framework and validation rules must be implemented to ensure the accuracy and reliability of information, especially for reporting and value-added services.
- **Financial Record Standards:** Digital records and books must be maintained according to generally accepted accounting principles.
- **Auditability:** An immutable audit trail must be implemented across all transactional and administrative actions to satisfy funder and regulatory audit requirements.

Data Utilization and Value-Adds

The platform captures data to support program operations, performance tracking, and potential value-added services for Agents.

- **Livelihood Tracking:** Data collected supports tracking the livelihood improvement of Koralink Agents, potentially including metrics on access to incentives like health insurance.
- **Recruitment Verification:** The system uses data generated on the platform (sales performance, transaction history) for micro-loan underwriting and relies on Data Agreements with key agent networks (MOMO, MTN, Equity) to cross-check income claims during the recruitment process.
- **Performance Metrics:** Data models capture critical metrics, including application scores, vulnerability assessments, transaction history, and stock order volumes, enabling robust performance tracking.
- **Geographic Data:** The platform is designed to handle location data via geofencing for monitoring Agent activity, which carries associated privacy considerations.

